

PortaMail

An Autonomous Mail Delivery Solution for Indoor Workspaces and Campuses



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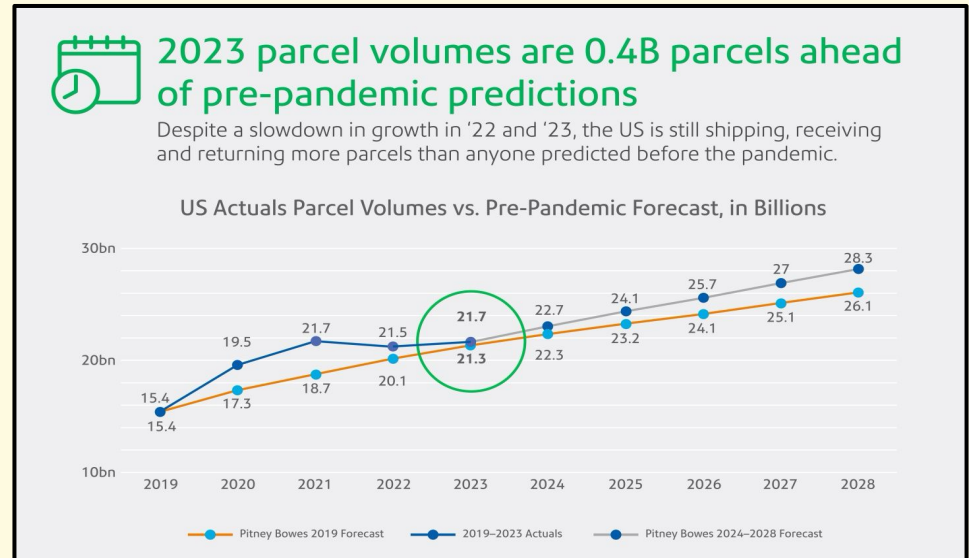
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Problem Statement

Surging parcel volumes and rising labor costs are creating significant mailroom bottlenecks, forcing employees to waste valuable time on internal deliveries and ultimately draining core staff productivity across offices, hospitals, and campuses

❏ “U.S. parcel volume reached **22.37 billion shipments in 2024**, a 95% increase since 2016.” — *Pitney Bowes Parcel Shipping Index (2024)*

❏ “The U.S. parcel market is projected to grow at an average of 5% per year, reaching **30.5 billion shipments by 2030**.” — *Nasdaq, citing Pitney Bowes Parcel Shipping Index (2024)*



Stakeholders

PortaMail Sponsor & Technical Advisor:
Dr. Logan Porter

SEBASTIAN MEJIAS – Project Manager

SIDDHARTH GANTI – Hardware Engineer

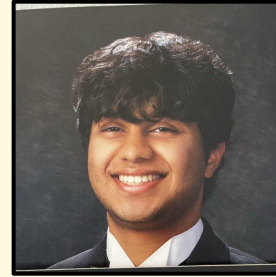
DAVID EZUMA – Software Engineer

PRANAY DHOOPAR – Mechanical Engineer

HENRY LOVELACE – Software Engineer



Sebastian Mejias



Siddharth Ganti



Henry Lovelace



David Ezuma

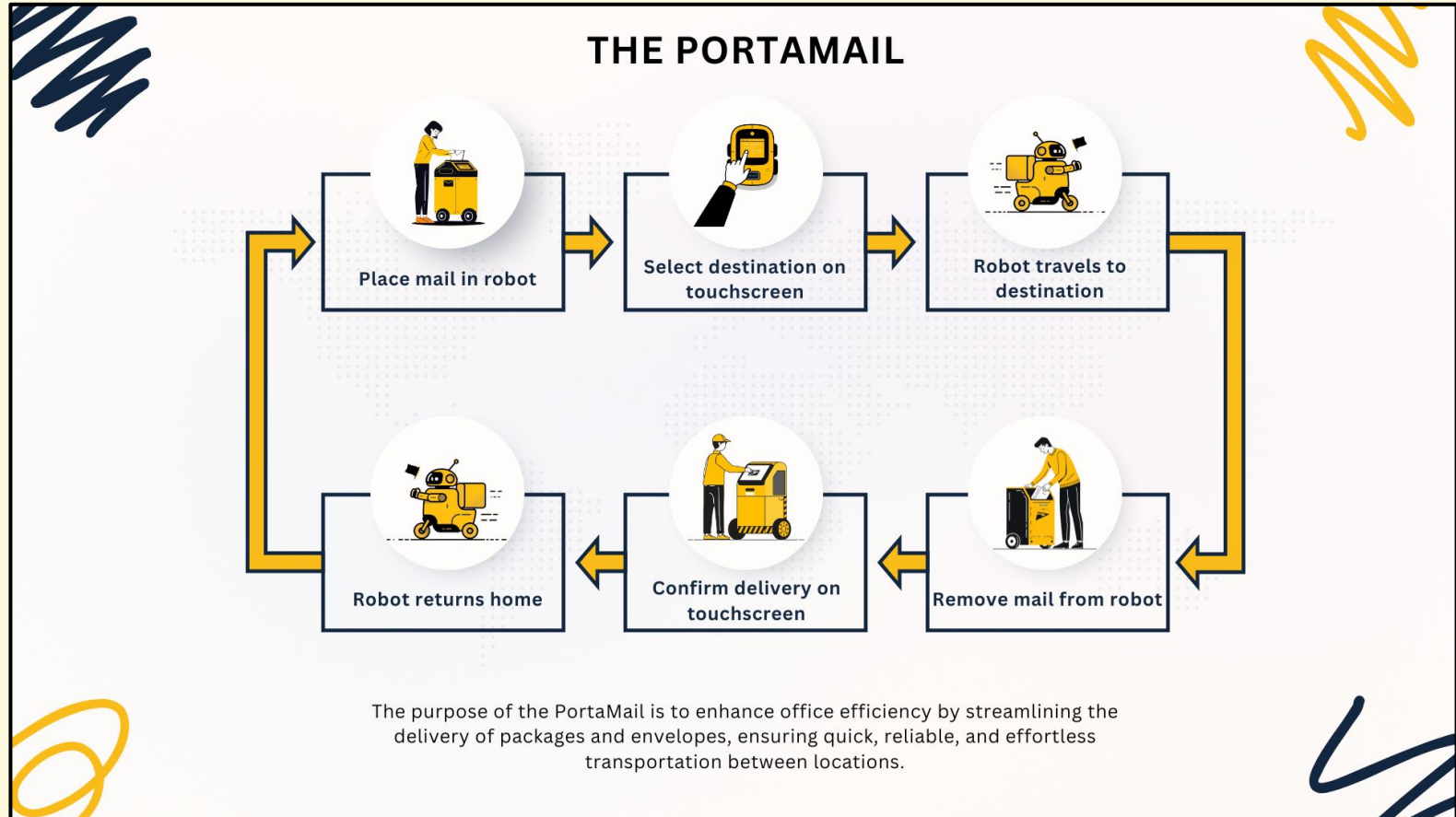


Pranay Dhoopar

Customers

User Type	Who They Are	What They Gain
Primary Users	Facilities & operations managers in large campuses	Reliable, automated delivery of small mail and packages across buildings
Secondary	Mailroom staff & internal logistics teams	Easier workload, fewer manual trips, improved tracking accuracy
Tertiary	Corporate, hospital, and university administration	Reduced labor costs, improved efficiency, and service consistency
End Beneficiaries	Employees, students, and office occupants	Faster deliveries, fewer lost items, and smoother operations

Conceptual Block Diagram



Project Scope

Large facilities face rising parcel volumes and inefficient, labor-intensive mail delivery.

In Scope

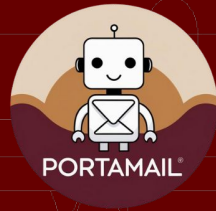
- Indoor navigation (hallways, mailrooms, offices)
- Autonomous route planning & obstacle avoidance
- Package handling ≤ 15 lbs
- User interface with touchscreen control
- Tracking and Notifications
- Power management for ≥ 2 hrs runtime

Out of Scope

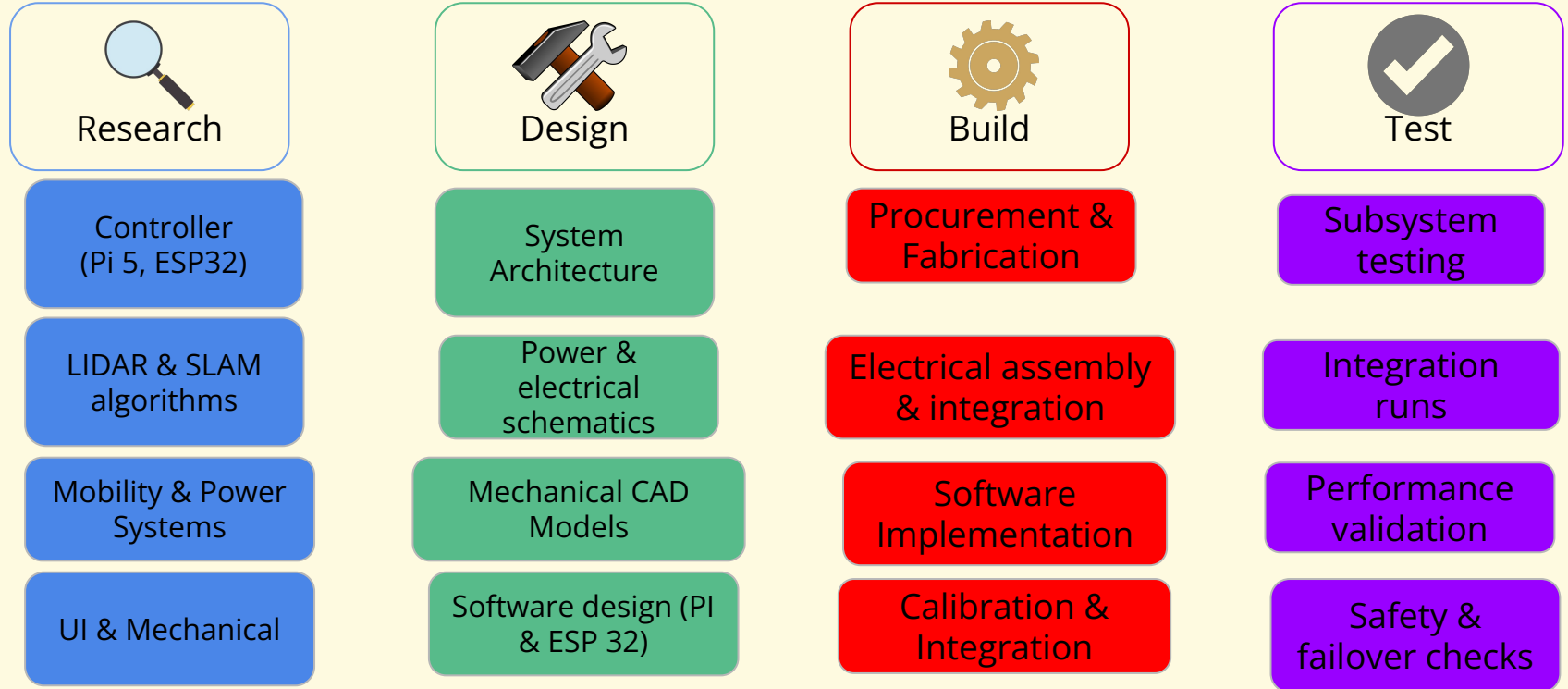
- Outdoor Delivery
- Stair or elevator navigation
- Heavy freight or large parcels
- Full warehouse automation
- Multi-robot coordination

Assumptions: Indoor only, <15 lbs packages, 1 prototype, yields to moving objects, healthy recipient

Work Breakdown Structure

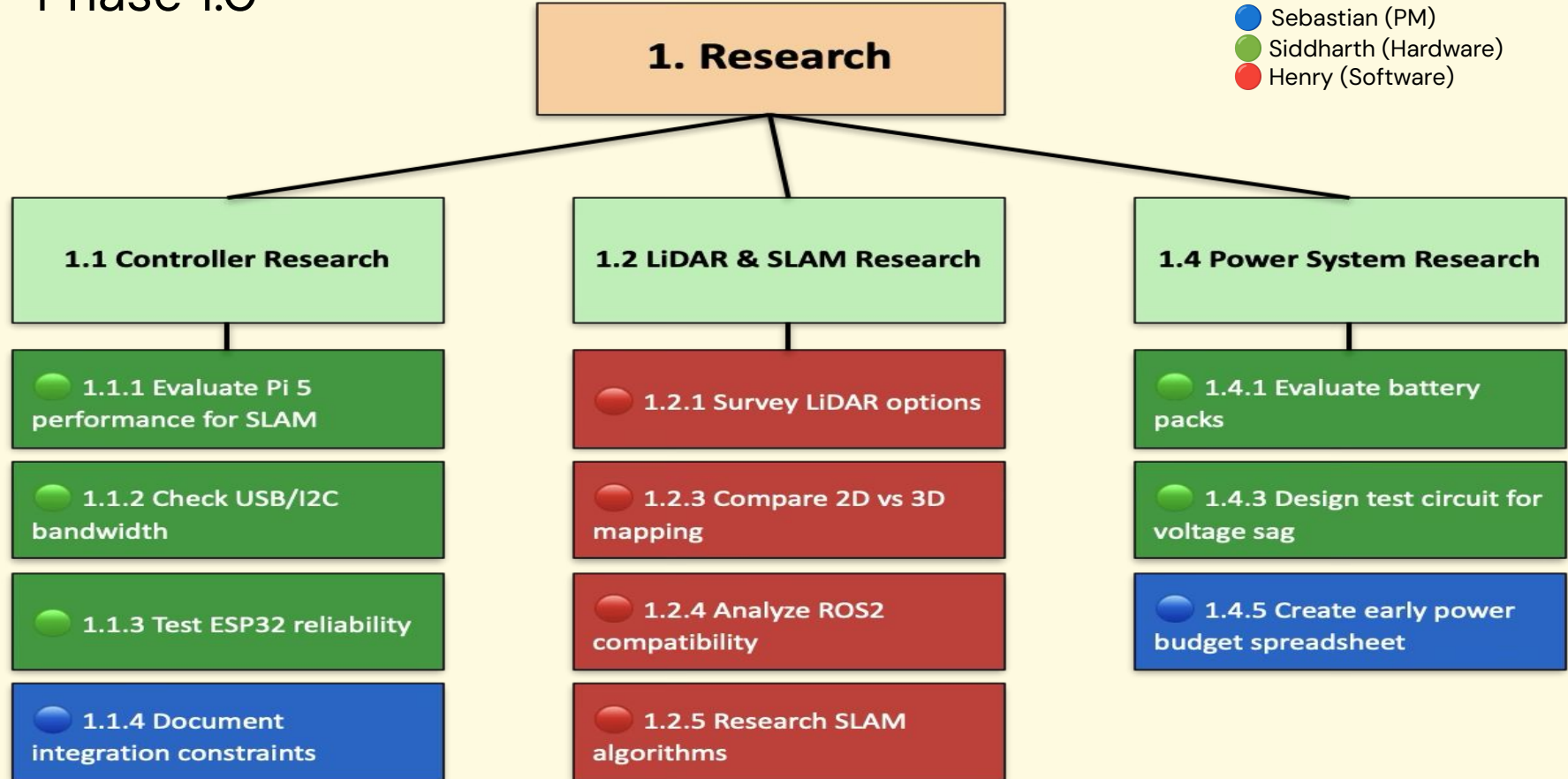


WBS High Level Overview



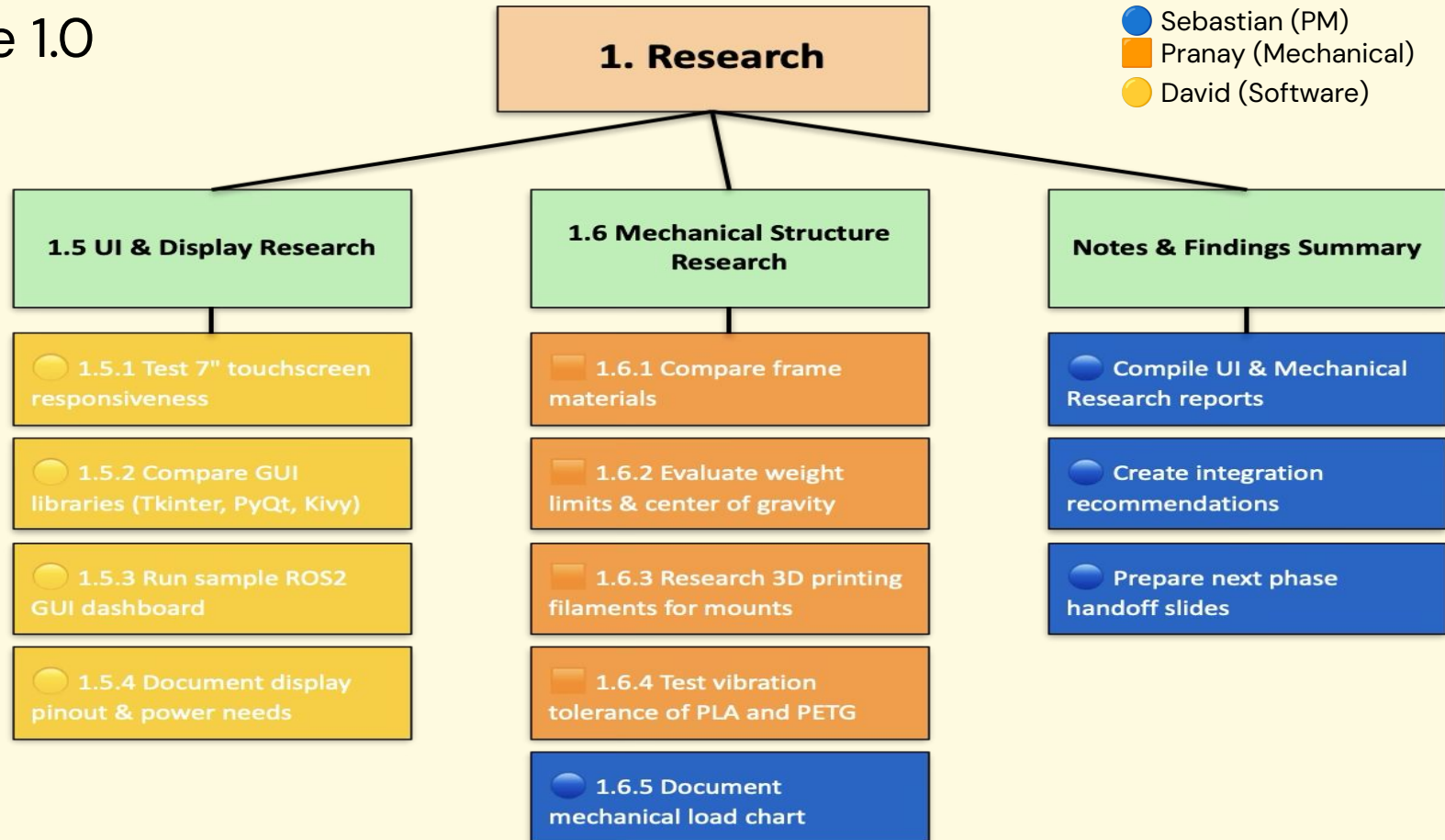
WBS Research

Phase 1.0



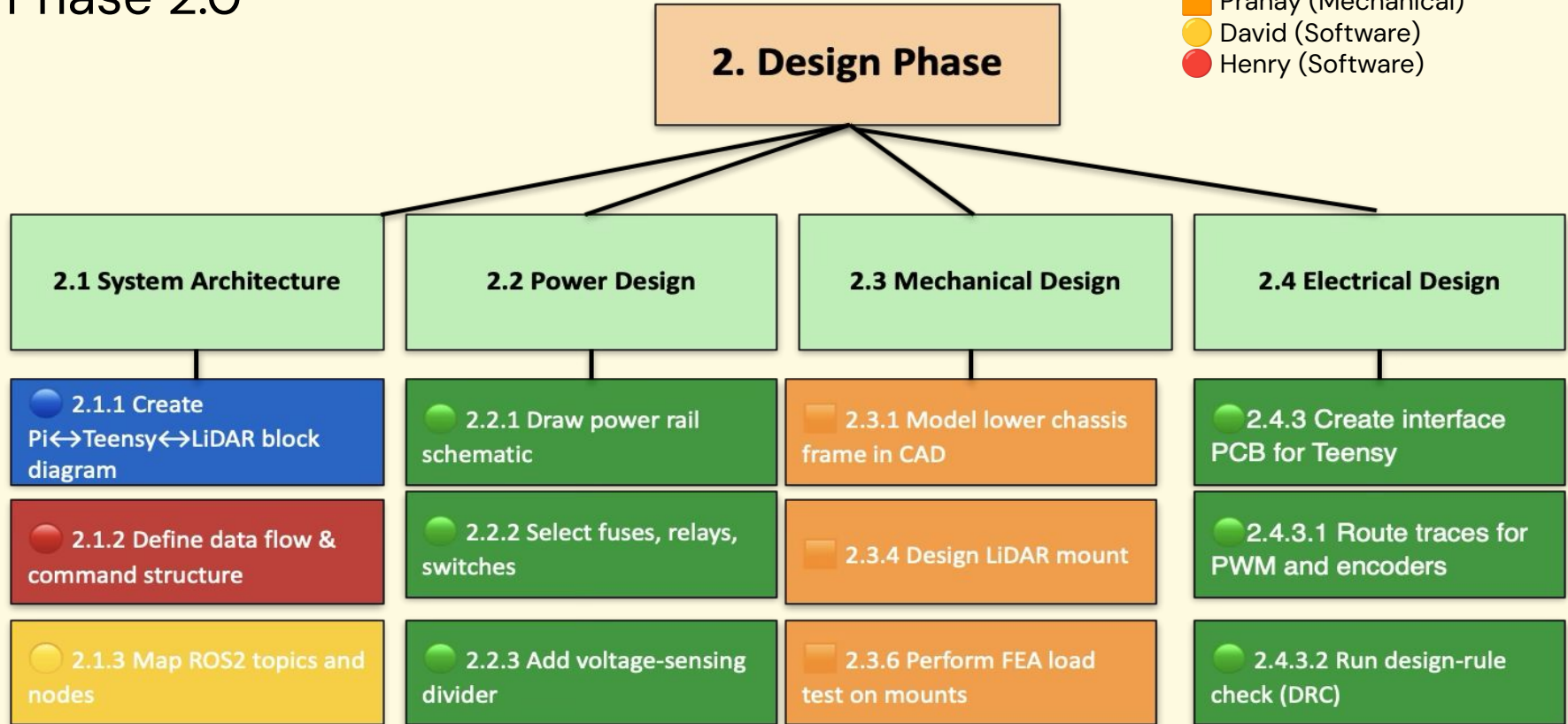
WBS Research

Phase 1.0



WBS Design Phase 2.0

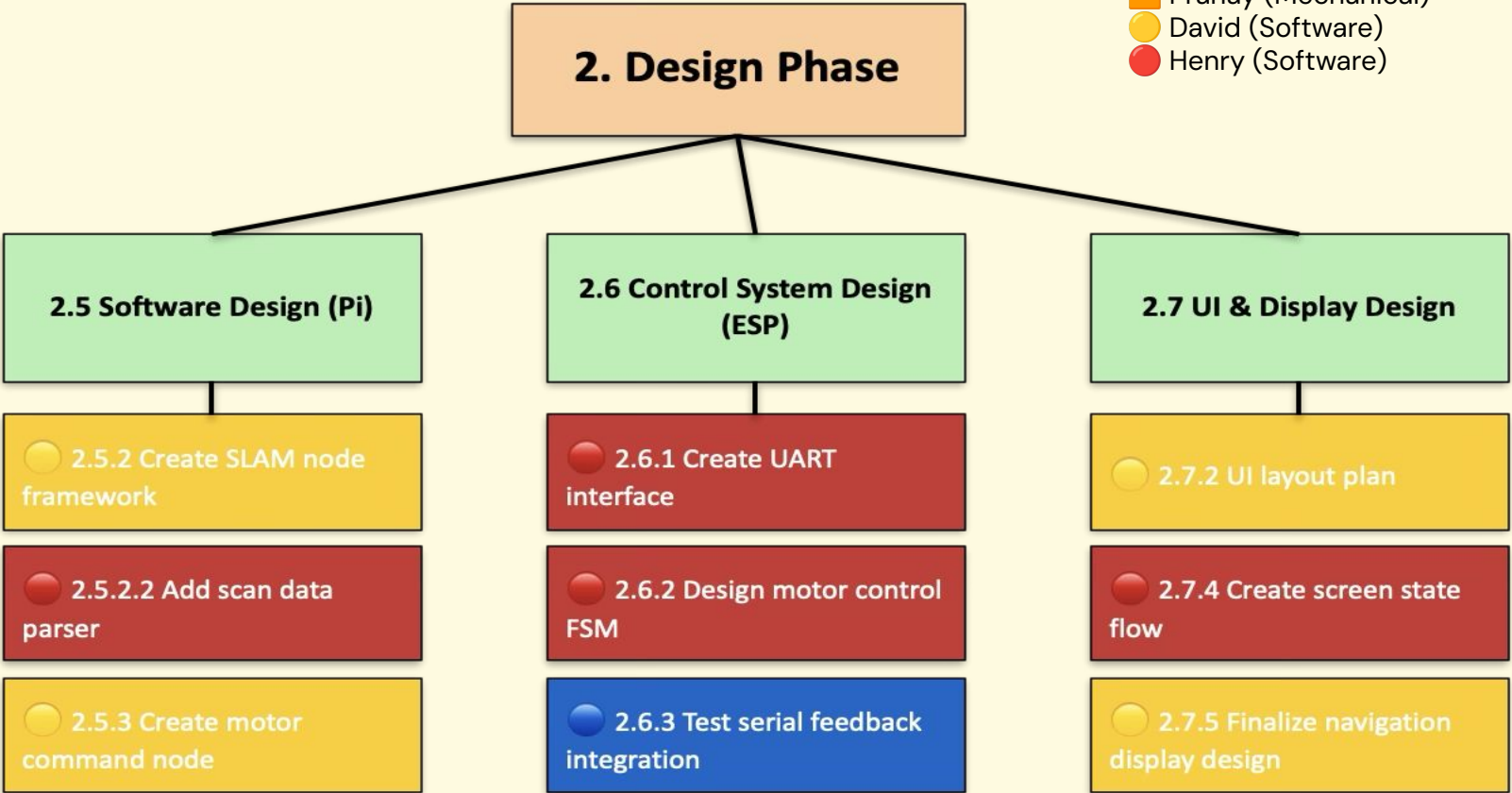
- Sebastian (PM)
- Siddharth (Hardware)
- Pranay (Mechanical)
- David (Software)
- Henry (Software)



WBS Design

Phase 2.0

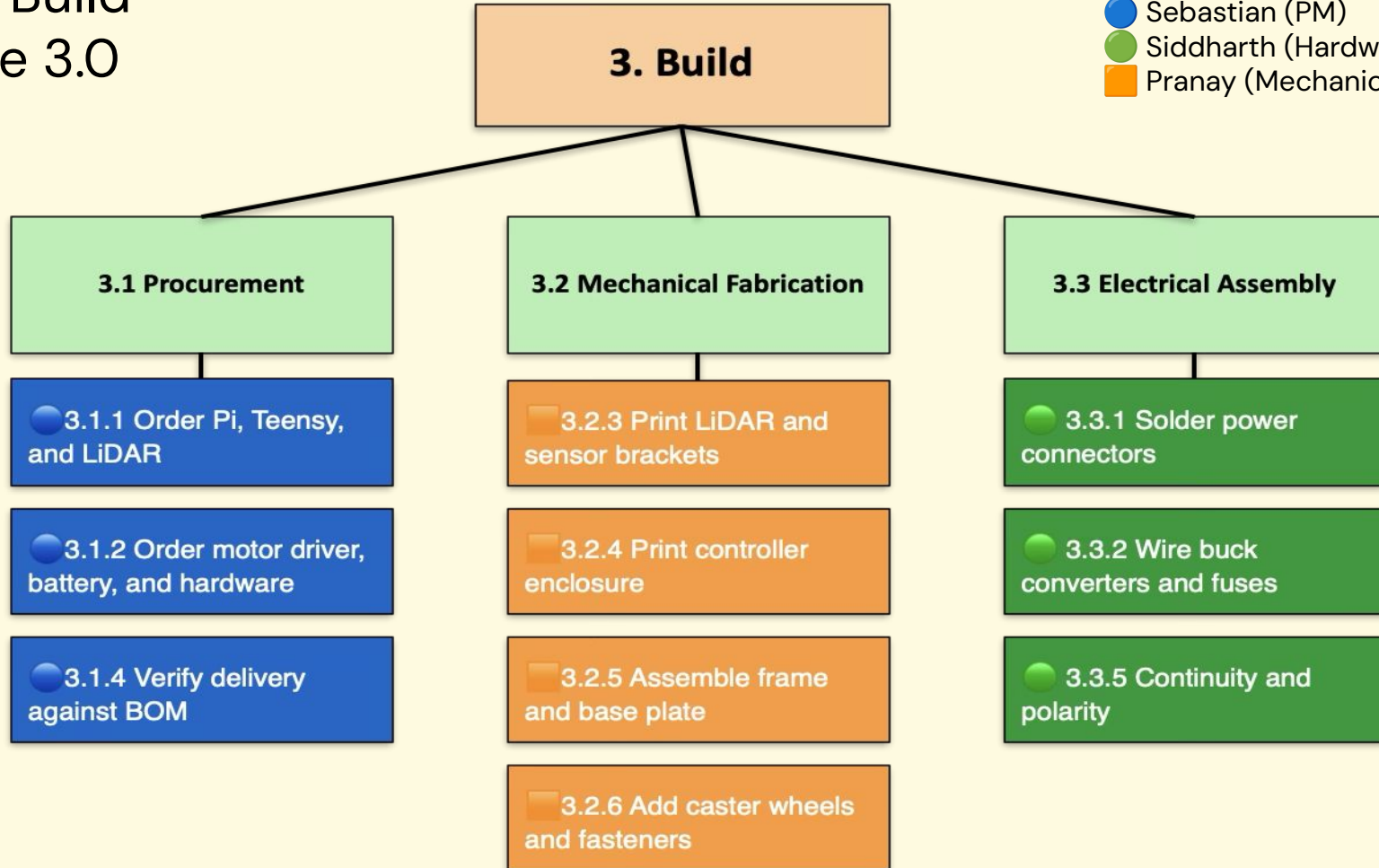
- Sebastian (PM)
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- Henry (Software)



WBS Build

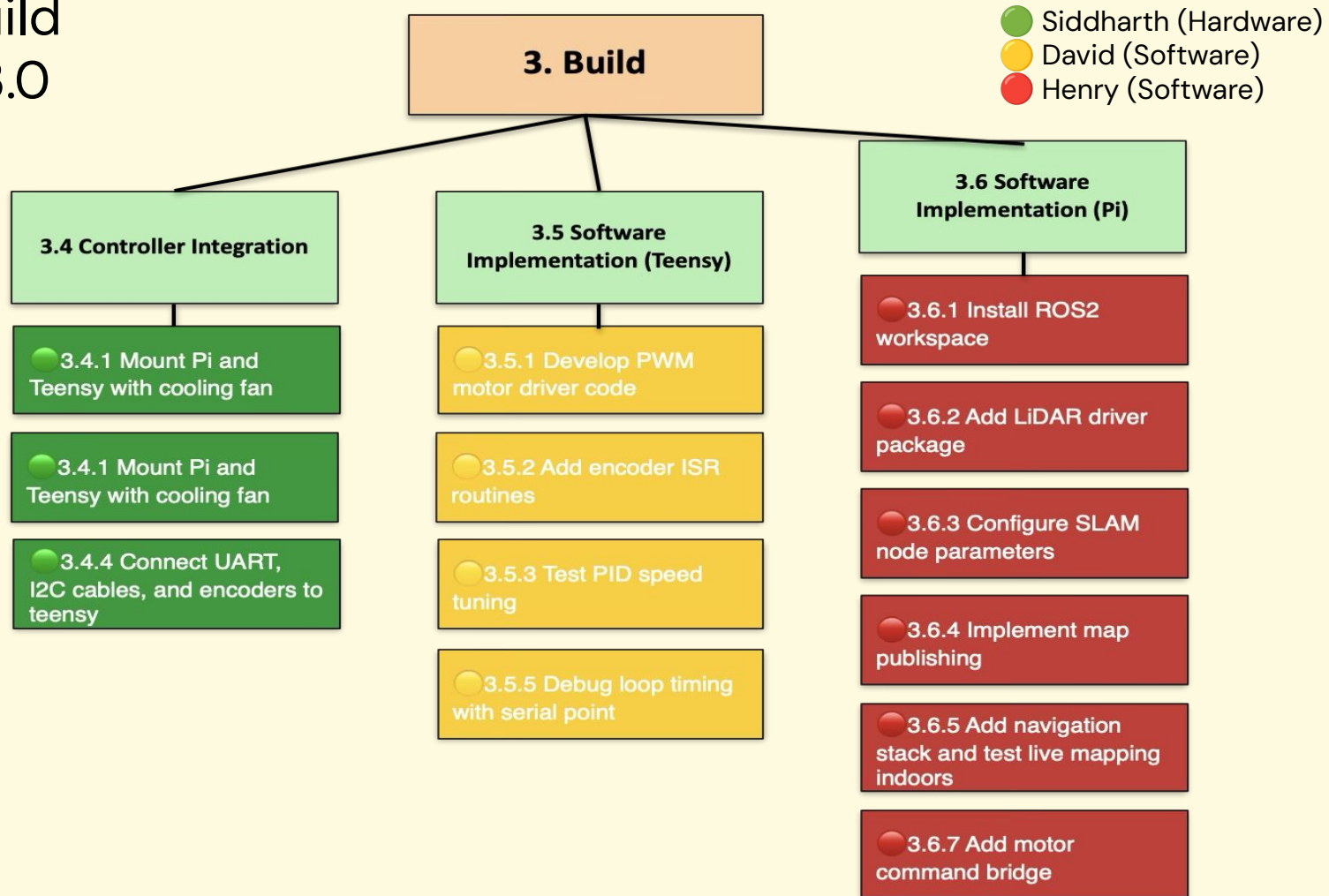
Phase 3.0

- Sebastian (PM)
- Siddharth (Hardware)
- Pranay (Mechanical)



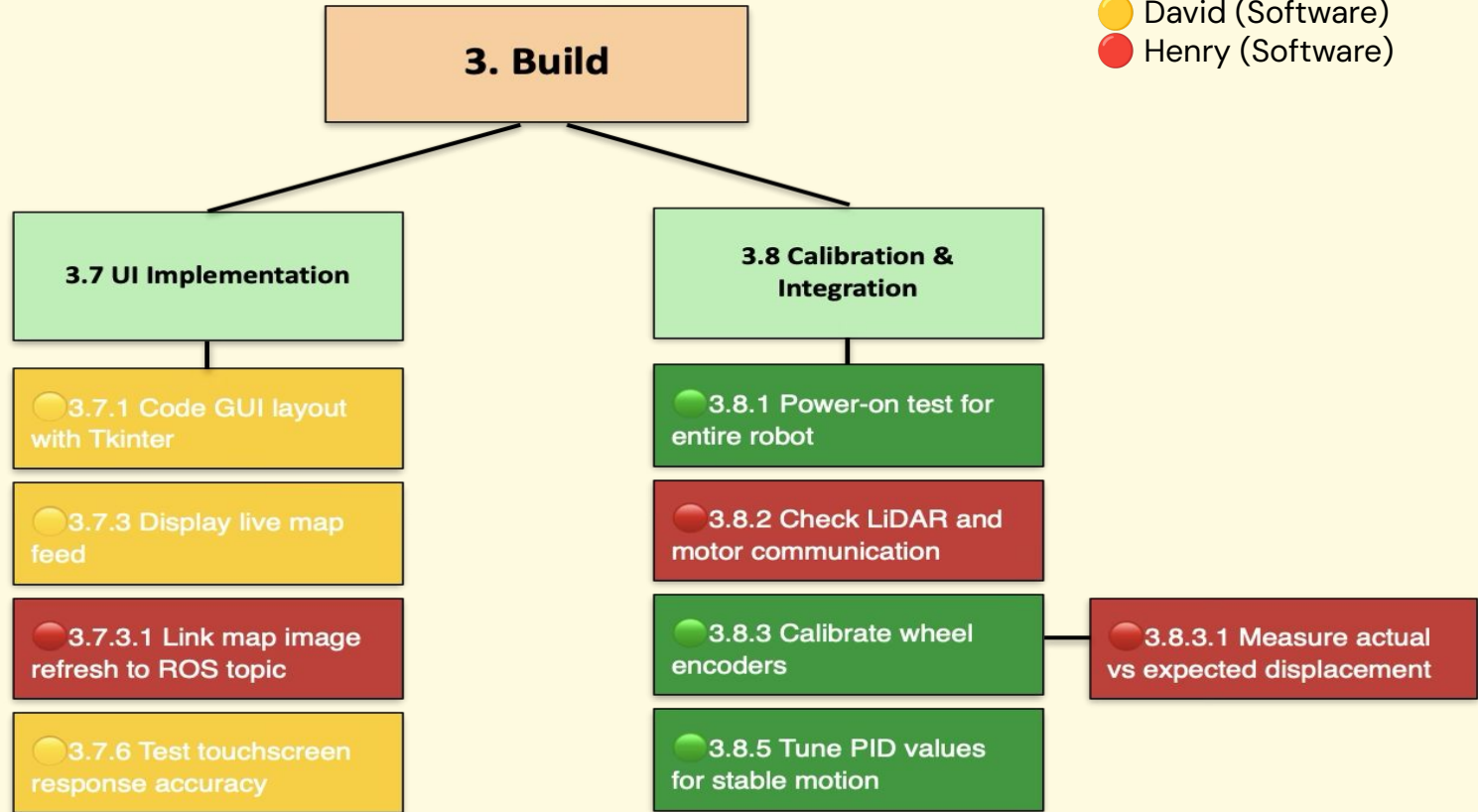
WBS Build

Phase 3.0



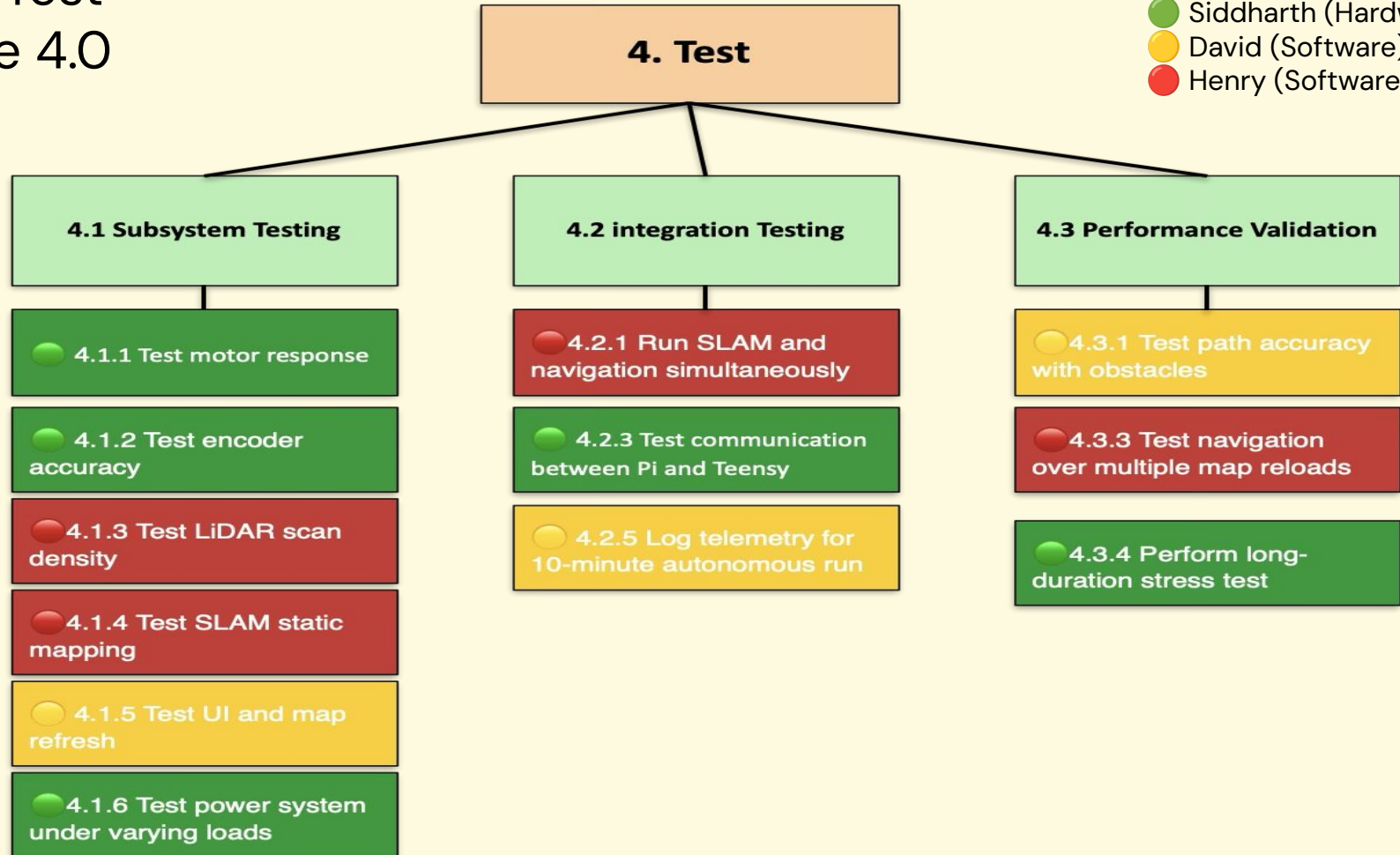
WBS Build Phase 3.0

- Siddharth (Hardware)
- David (Software)
- Henry (Software)

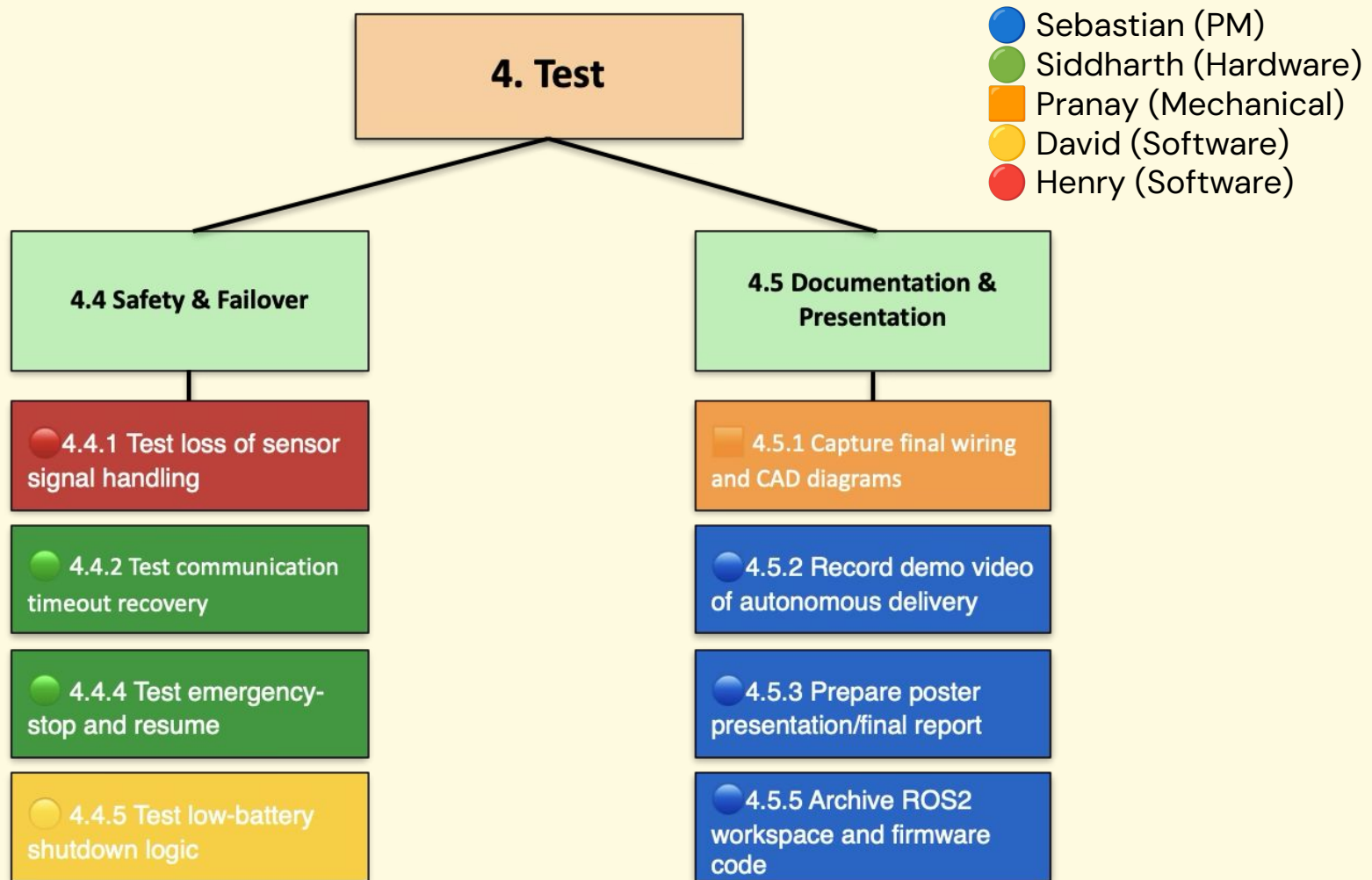


WBS Test Phase 4.0

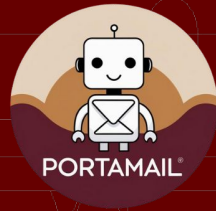
- Siddharth (Hardware)
- David (Software)
- Henry (Software)

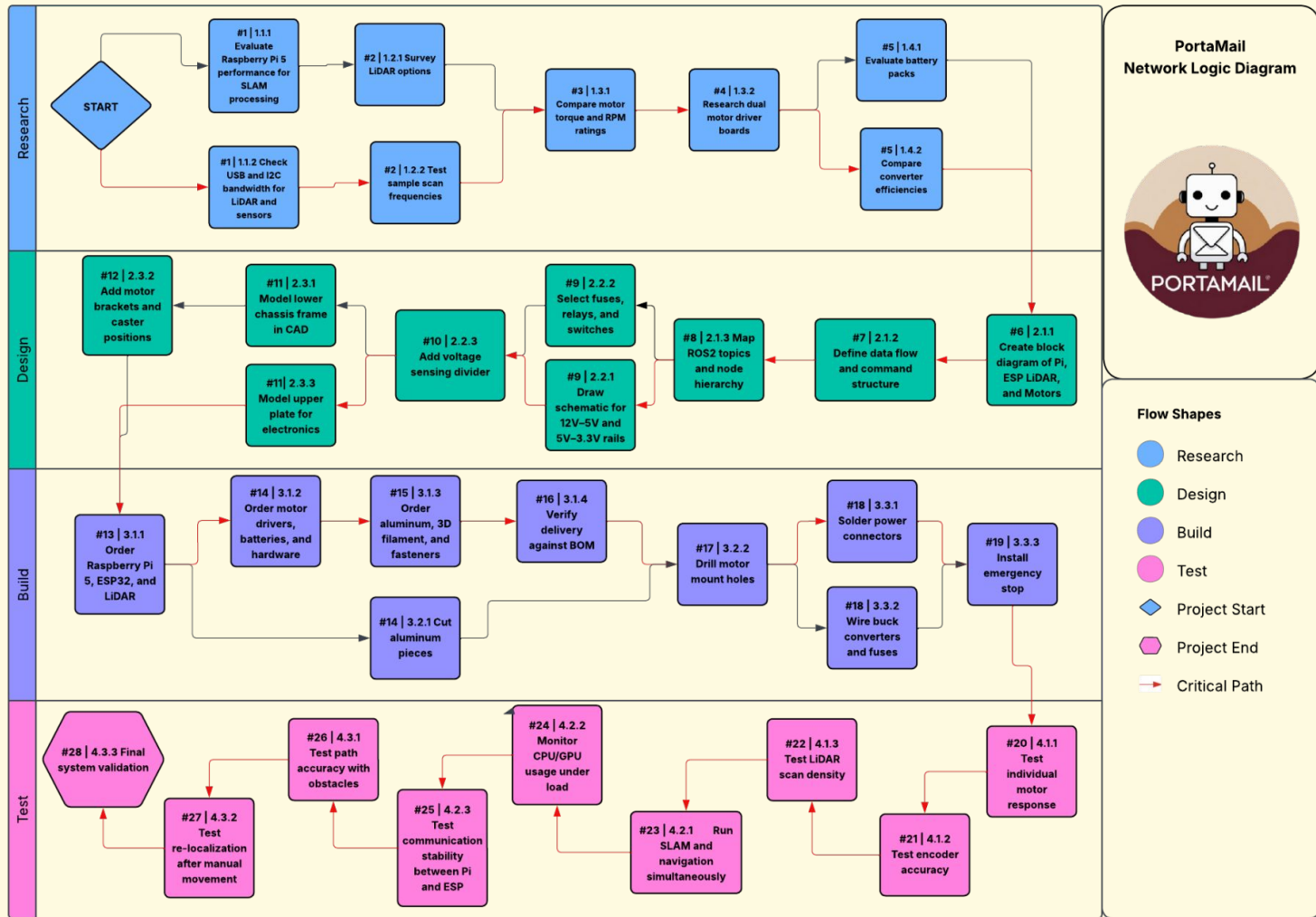


WBS Test Phase 4.0



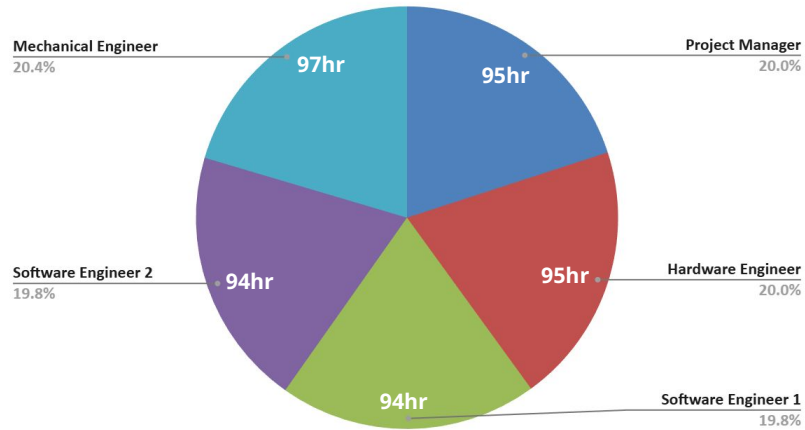
Network Logic Diagram



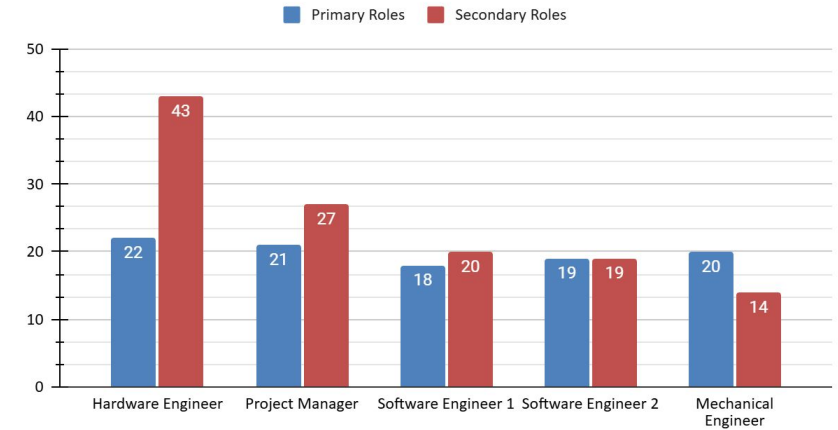


Resource Plan

Total Hours Per Person



Primary Roles and Secondary Roles



Milestones

Component Research & Specification Review

Research available LiDAR systems evaluate scanning range, refresh rate, and data transfer speed to determine best fit for indoor navigation and SLAM reliability.

Project Scope Acceptance Sheet

Finalize and sign the team's project scope agreement with defined deliverables, constraints, and measurable success criteria.

PDR Presentation Submission

Submit finalized PDR presentation and record individual presentations. Incorporate instructor and peer feedback into revisions.

FTP & Peer Evaluation

Submit the complete project documentation (CAD files, FBDs, software maps, risk matrix, and budget). Complete peer evaluations by deadline.

Demo #2 – System Optimization

Test full integration between motor feedback loop, SLAM navigation, and LCD interface for user status monitoring.

10/13/2025

10/17/2025

10/20/2025

10/27/2025

11/04/2025

12/01/2025

12/08/2025

01/27/2026

03/31/2026

04/28/2026

CAD Model & System Layout Design

Develop a complete CAD model of the robot chassis, showing motor placement, battery housing, LiDAR mount, and control board positions. Ensure all components fit securely and maintain proper weight balance.

PDR Presentation Development

Prepare the Preliminary Design Review slides including system overview, functional diagrams, CAD model, risks, mitigation strategies, and budget summary.

Prototype Software & Hardware Demonstration

Deliver partial software demonstration showing ROS-based LiDAR data visualization, motor control test, or simulated navigation environment.

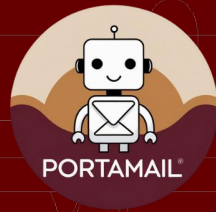
Demo #1 – Autonomous Navigation

Demonstrate basic navigation using SLAM map and manual input start. Confirms ROS communication between microcontrollers and LiDAR integration.

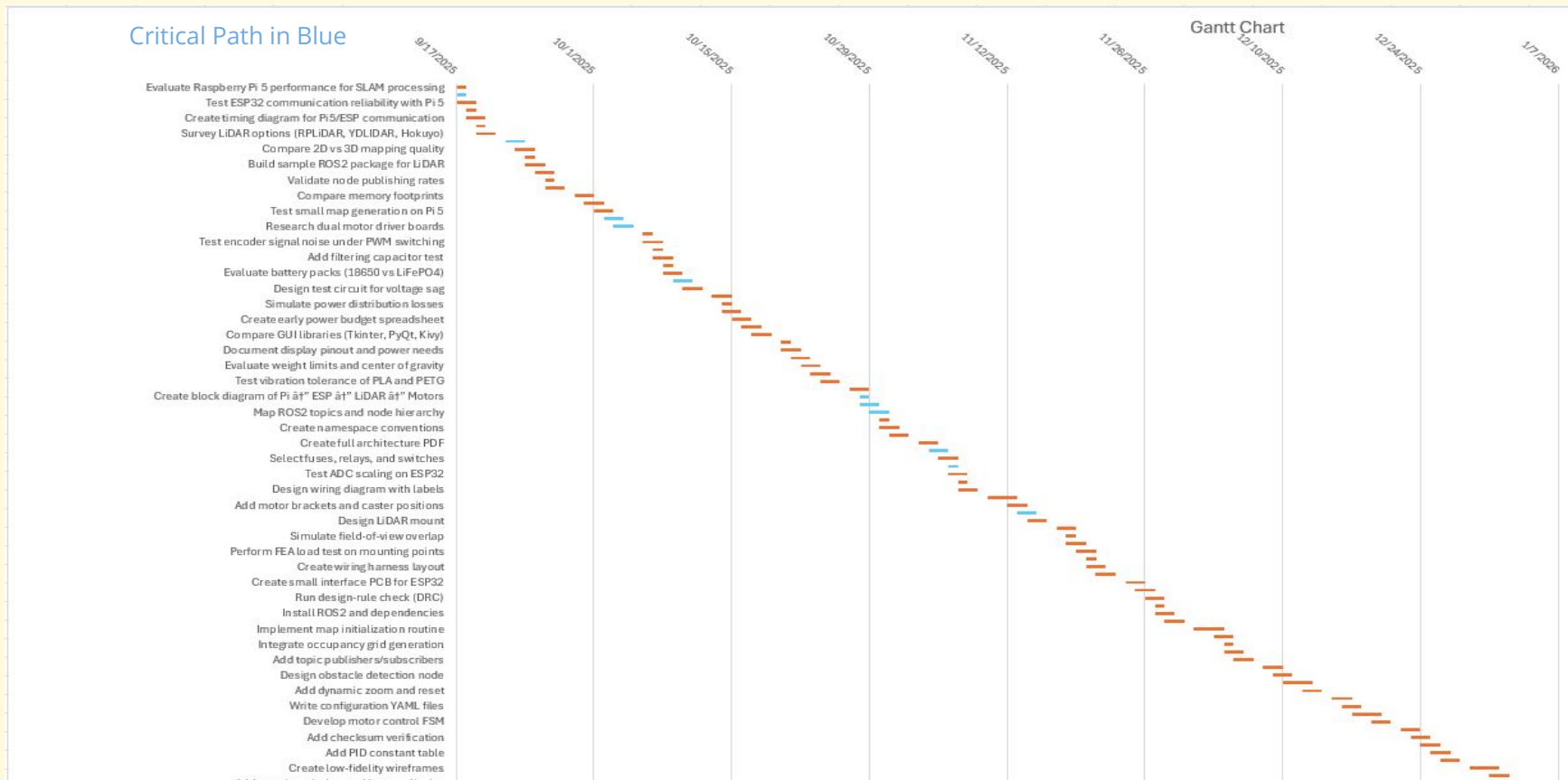
Final Demo

Showcase end-to-end operation from mail input to successful delivery and return-to-base routine with all safety features active.

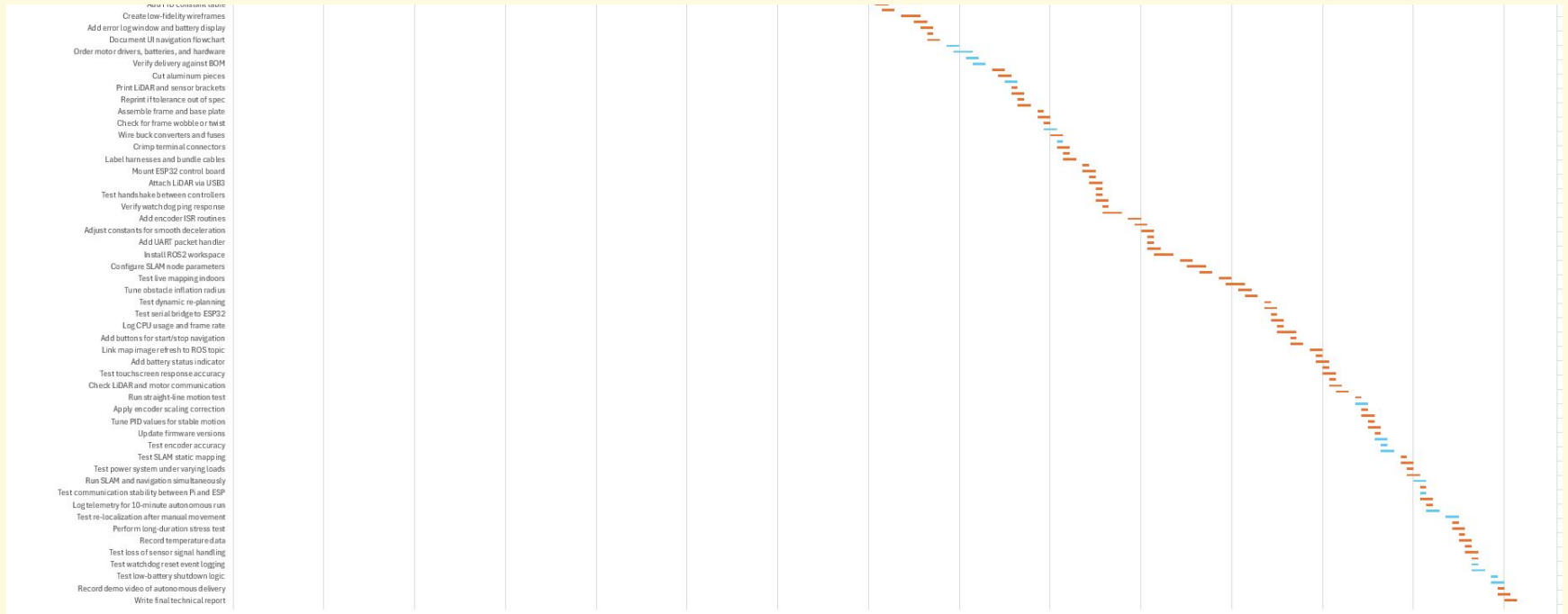
Gantt Chart



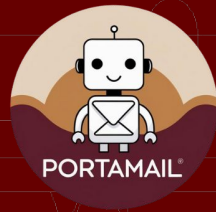
Gantt Chart



Gantt Chart



Risks and Mitigation



Risks

Mechanical failure
of wheel or motor
assembly

Team member
availability/conflict

SLAM accuracy
degradation due to
poor sensor data

Budget goes over
due to hardware
replacement

Software integration
delay (SLAM + motor
control)

Overheating of
Raspberry Pi 5, motor
drivers or battery

Component
shortage or
shipping delays

Robot hits a
person or an
object

UI touchscreen
malfunction

Risk Severity

Impact Level

3

4

5

Probability Level

2

3

4

Mechanical failure
of wheel or motor
assembly

Overheating of
Raspberry Pi 5, motor
drivers or battery

Robot hits a person
or an object

UI touchscreen
malfunction

Team member
availability/conflict

Budget goes over
due to hardware
replacement

Software integration
delay (SLAM + motor
control)

Component
shortage or
shipping delays

SLAM accuracy
degradation due to
poor sensor data

Mitigation Strategies

Mechanical failure
of wheel or motor
assembly

Use quality motor drivers, stress test the chassis and joints and keep spare components on hand

Budget goes over
due to hardware
replacement

Track expenses weekly, maintain spare funding

Component
shortage or
shipping delays

Order early, identify backup vendors; prioritize long lead items

Mitigation Strategies

Team member
availability/conflict

Use shared calendar, keep communicating, hold people responsible

Software integration
delay (SLAM + motor
control)

Use ROS2 nodes and integrate incrementally

Robot hits a
person or an
object

Emergency stop switch and limit motor speed indoors

Mitigation Strategies

SLAM accuracy
degradation due to
poor sensor data

Calibrate sensors, possibly use filtering, make sure to have a good map before runs

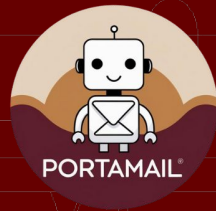
Overheating of
Raspberry Pi 5, motor
drivers or battery

Add heatsinks and active cooling

UI touchscreen
malfunction

Test UI firmware early, maybe add a serial interface

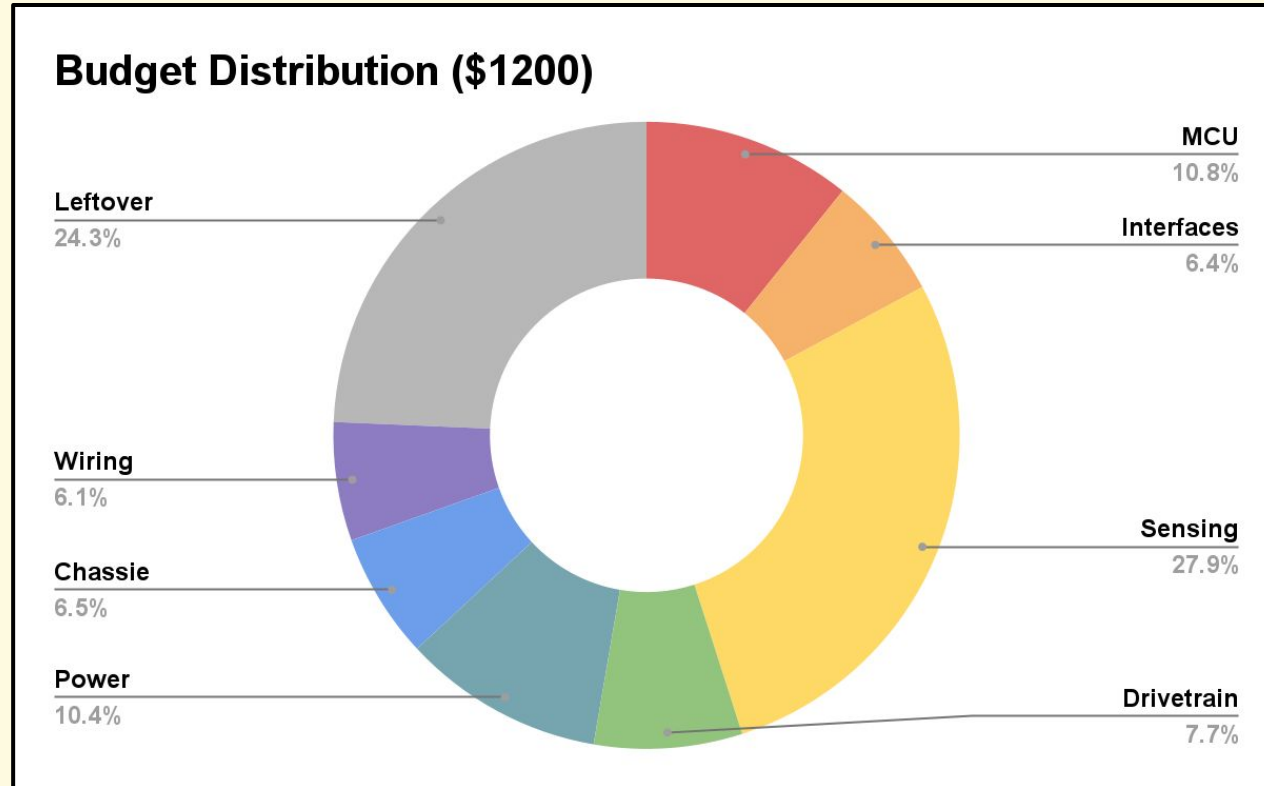
Budget



Budget: Material Cost

Main Items					
Category	Part	Model & Link	Price	Quantity	Total
MCU	High Level Controller	Raspberry Pi 5 8GB	\$80.00	1	\$80.00
MCU	Low Level Controller	Teensy 4.0	\$23.80	1	\$23.80
UI	Pi Touchscreen	SC1227	\$60.00	1	\$60.00
Sensing	LiDAR	RPLIDAR A2M12	\$209.00	1	\$209.00
Sensing	Camera	RPi Camera	\$37.90	1	\$37.90
Sensing	UltraSonic Ranger	101020010	\$4.30	4	\$17.20
Sensing	IMU	Adafruit 2472	\$34.95	1	\$34.95
Drivetrain	Motor Driver	4 Pack L298N	\$20.00	1	\$20.00
Drivetrain	Motor with Encoder	FIT0186	\$16.50	2	\$33.00
Power	Battery	Talencell 26400mAh	\$63.99	1	\$63.99
Other Items					\$261.05
Net Total	\$840.89		Total + Tax (8%)		\$908.16

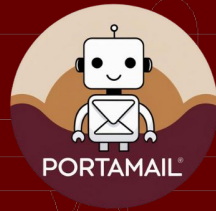
Budget: Material Cost



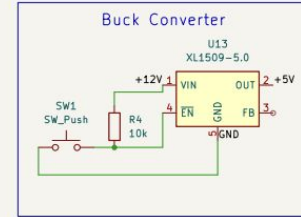
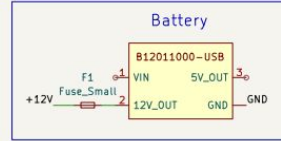
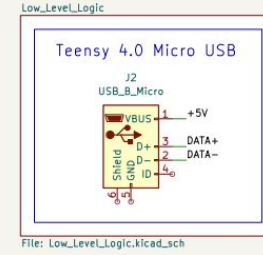
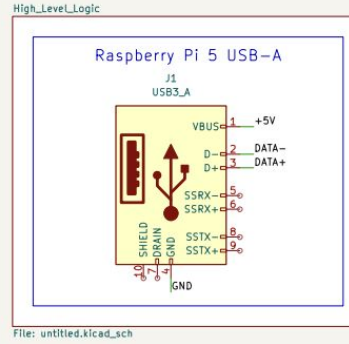
Budget: Resource Costs

Resource Costs				
Title	Yearly Salary	Hourly Rate	Billable Hours	Amount Owed
Project manager	\$80,000.00	\$38.50	95	\$3,657.50
Mechanical Engineer	\$75,000.00	\$36.00	97	\$3,492.00
Software Engineer 1	\$82,000.00	\$39.50	94	\$3,713.00
Software Engineer 2	\$82,000.00	\$39.50	94	\$3,713.00
Hardware Engineer	\$78,000.00	\$37.50	95	\$3,562.50
			Total Amount	\$18,138.00

Functional Block Diagram



FBD: Root Logic



Raspberry Pi 5 and Teensy 4.0 connect to each other
This is the root schematic that shows how the

Sheet: /
File: capstone_pl.kicad_sch

Title: PortaMail Root Shcematic

Size: A4 Date: 2025-10-29

KiCad E.D.A. 9.0.0

Rev: 0

Id: 1/3

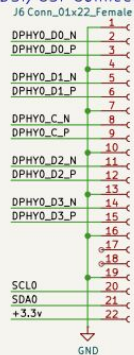
FBD: High Level Logic

Raspberry Pi 5

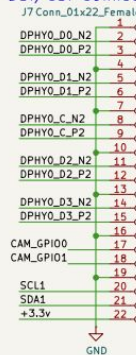
Fan Connector



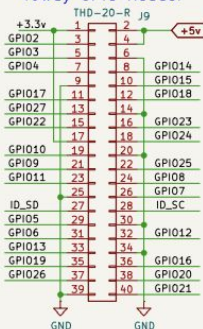
DSI/CSI Connector #1



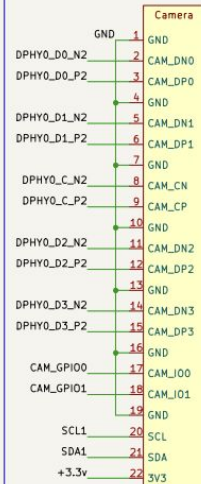
DSI/CSI Connector #2



40Way GPIO Header



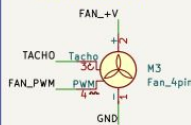
Camera Module



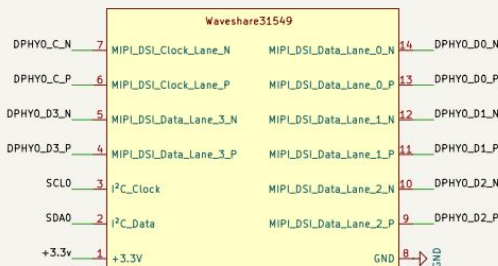
LIDAR



Active Cooler



Touchscreen LCD



This shows the schematics of what will be connected to and controlled by the Raspberry Pi 5

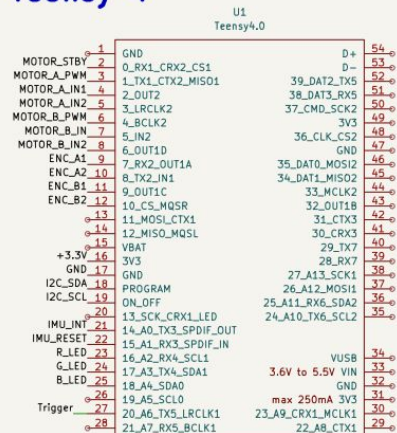
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File: untitled.kicad_sch

Title: Raspberry Pi 5 - High Level Logic Schematic

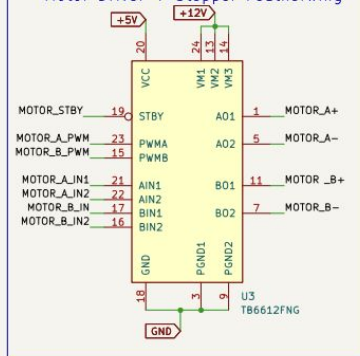
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FBD: Low Level Logic

Teensy 4



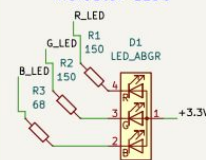
Motor Driver + Stepper Featherwing



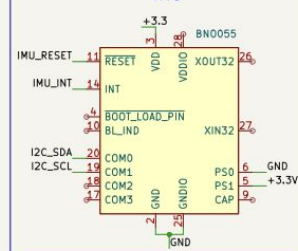
Ultrasonic Ranger



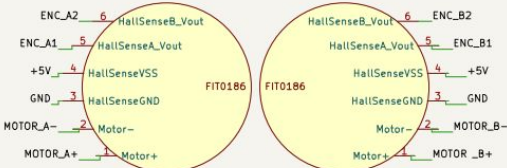
Indicator LEDs



IMU



2 Motors & Encoders



This shows the schematics of what will be connected to and controlled by the Teensy 4.0

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File: Low_Level_Logic.kicad_sch

Title: Teensy 4.0 - Low Level Logic Schematic

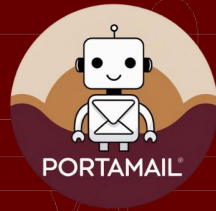
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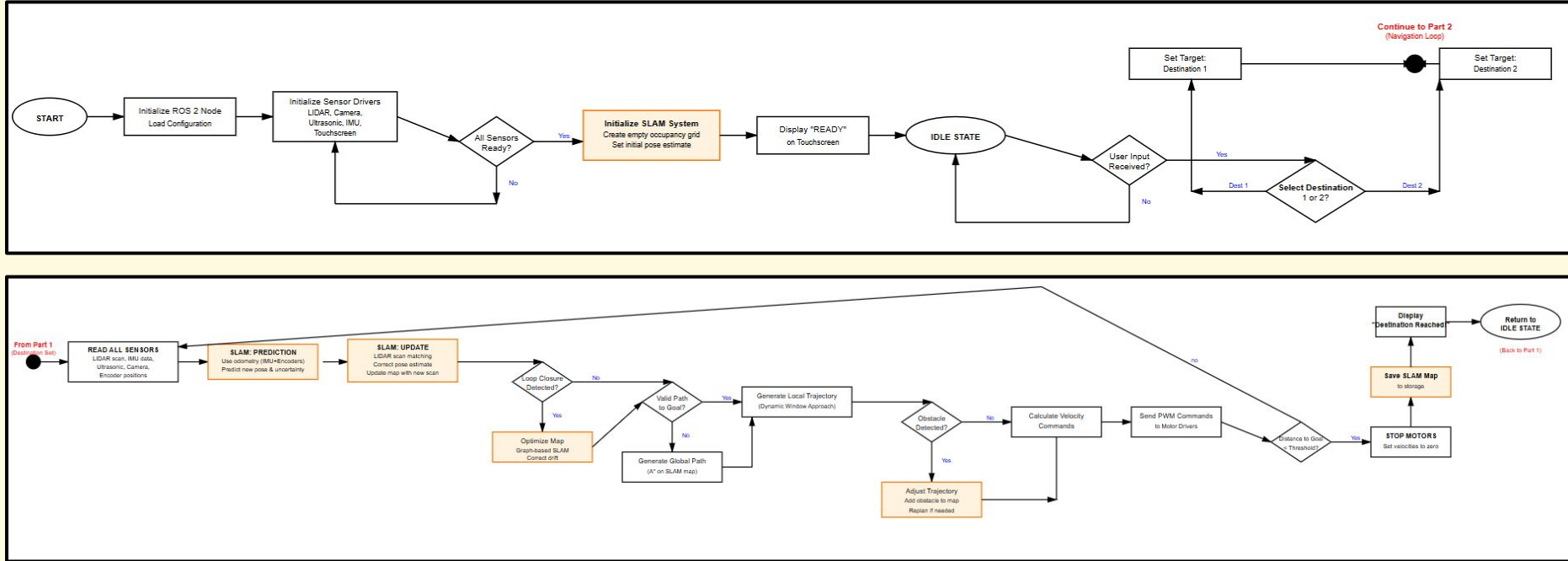
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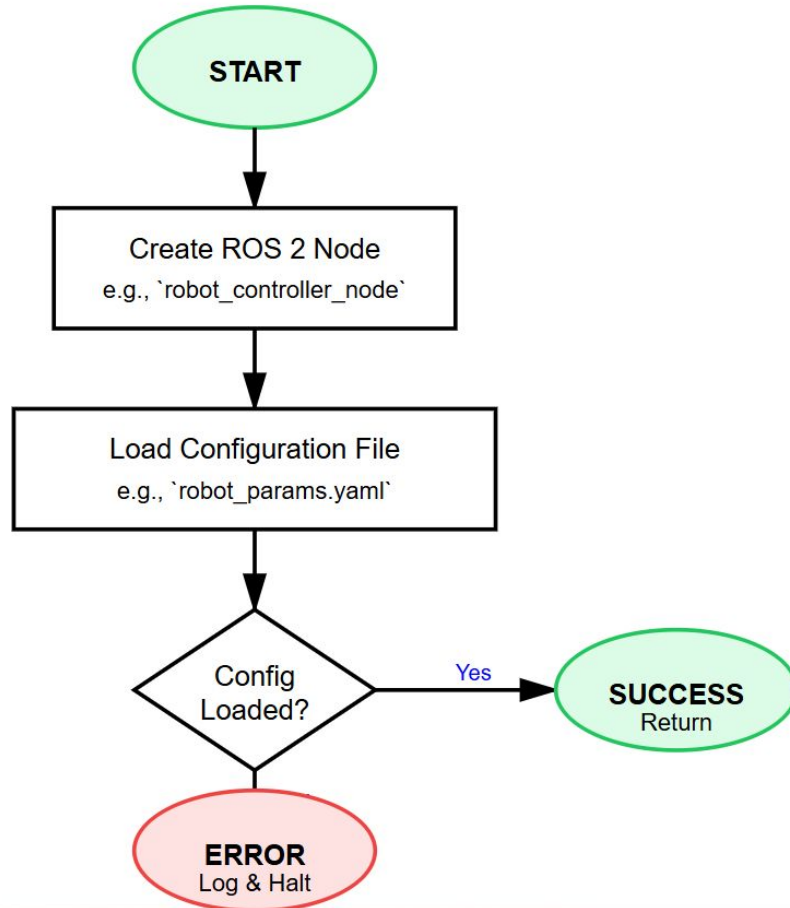
Software Process Map



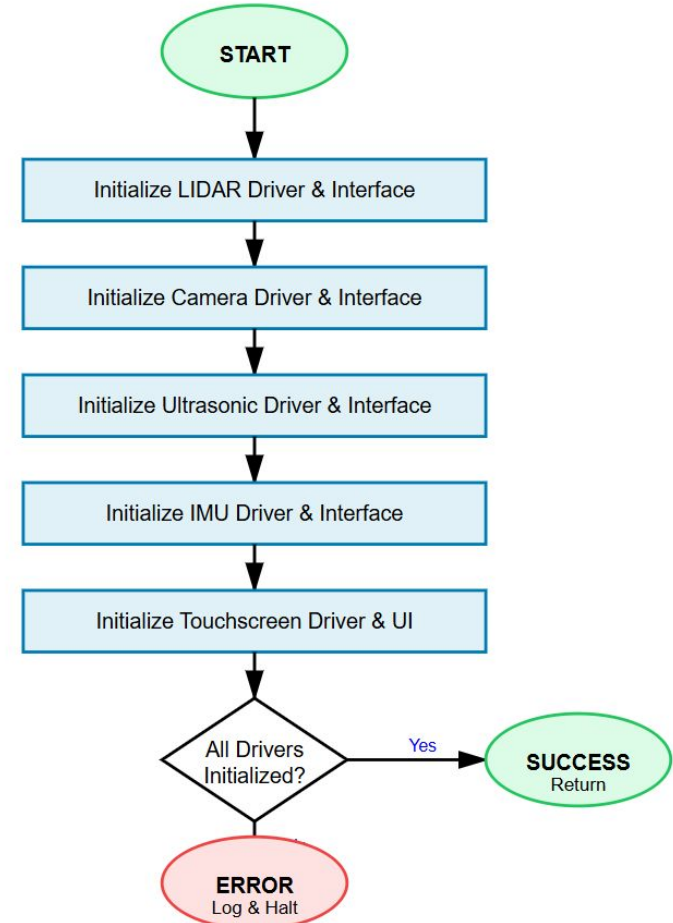
Software Process Map



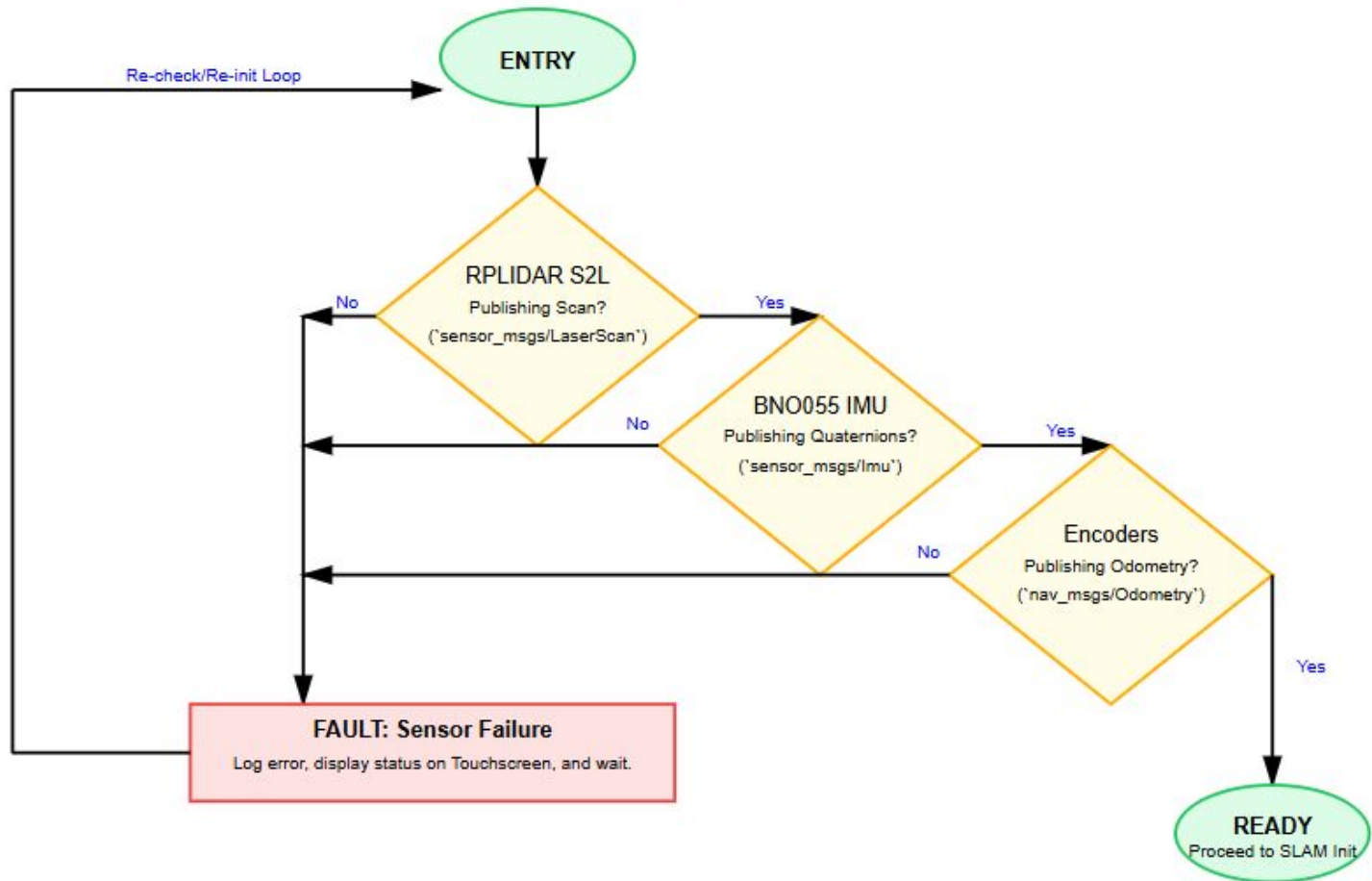
Sub-Flow: Initialize ROS 2 Node / Load Configuration



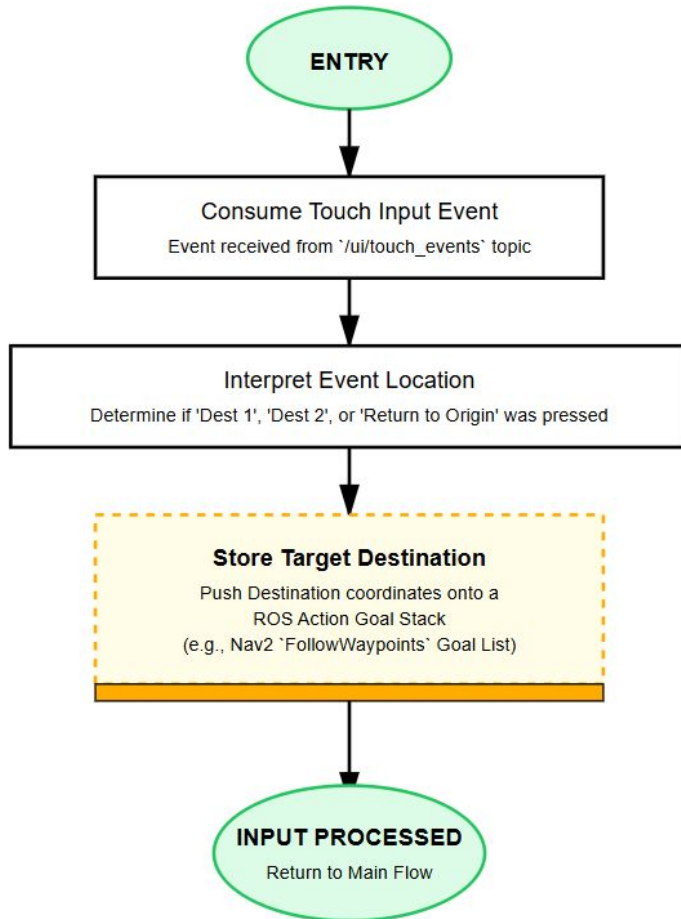
Sub-Flow: Initialize Sensor Drivers



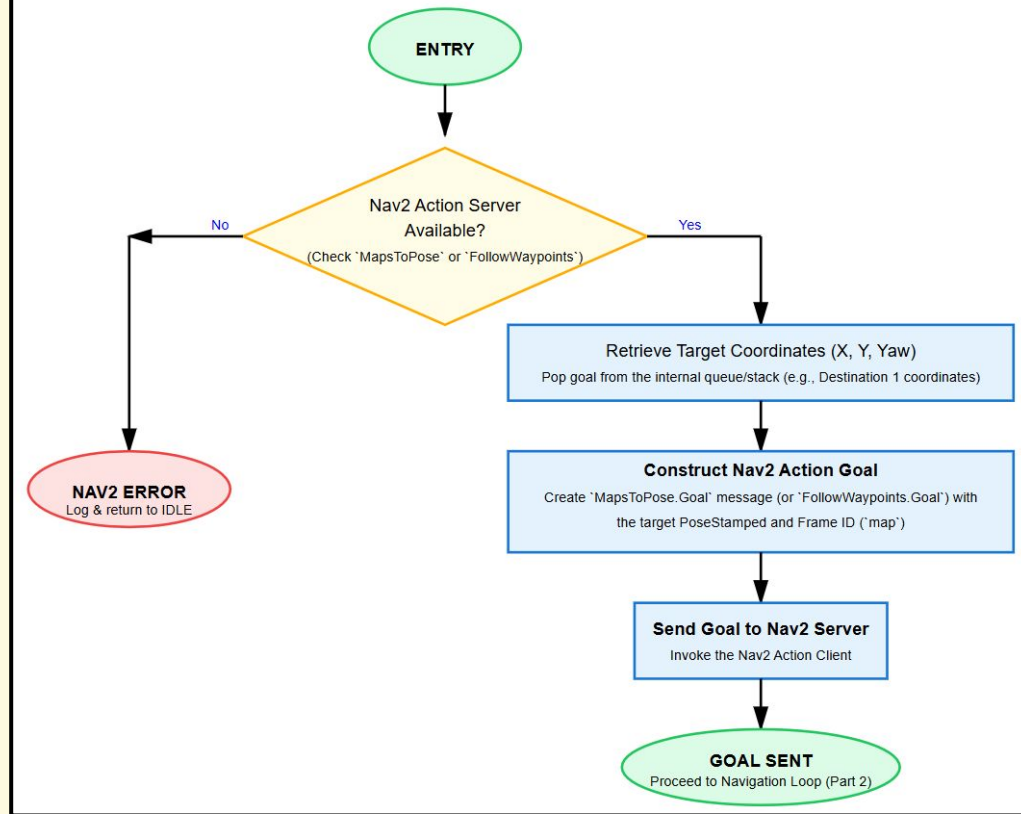
Sub-Flow: All Sensors Ready?



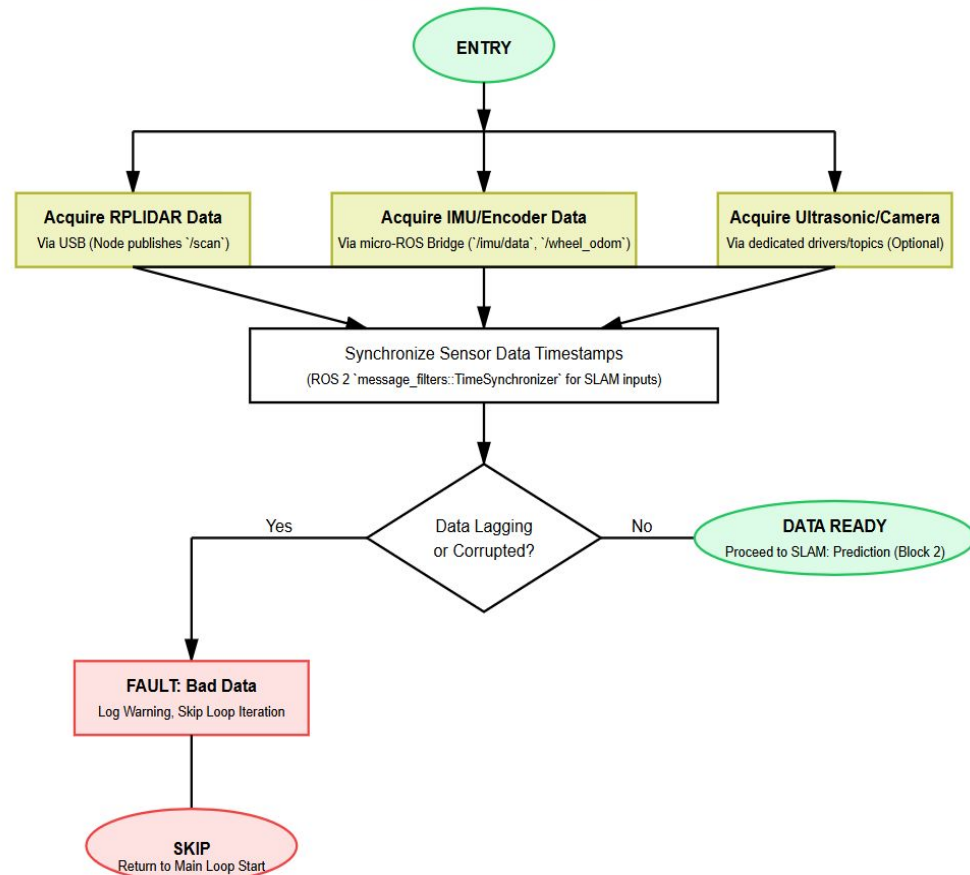
Sub-Flow: User Input Received & Processed



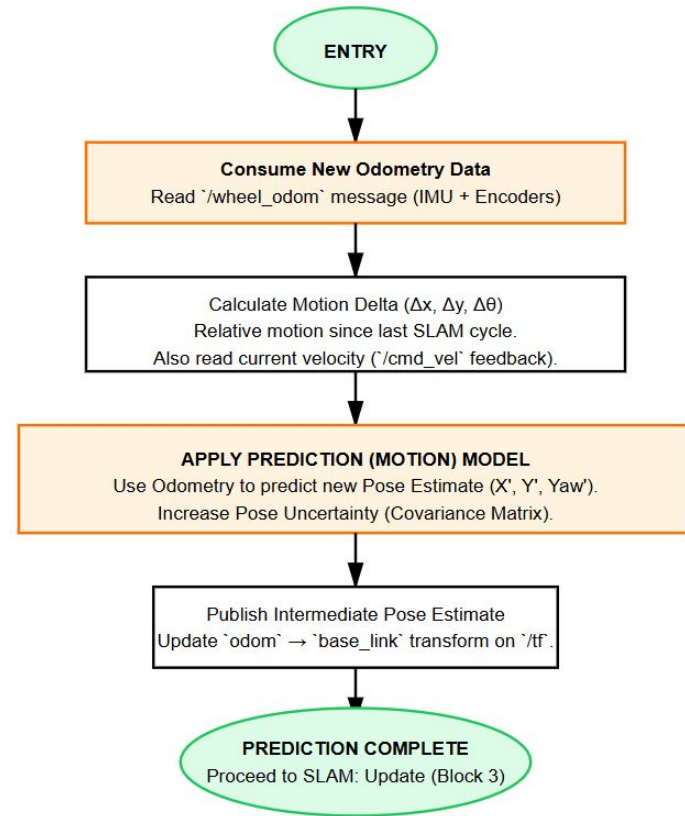
Sub-Flow: Set Target (Navigation Goal Preparation)



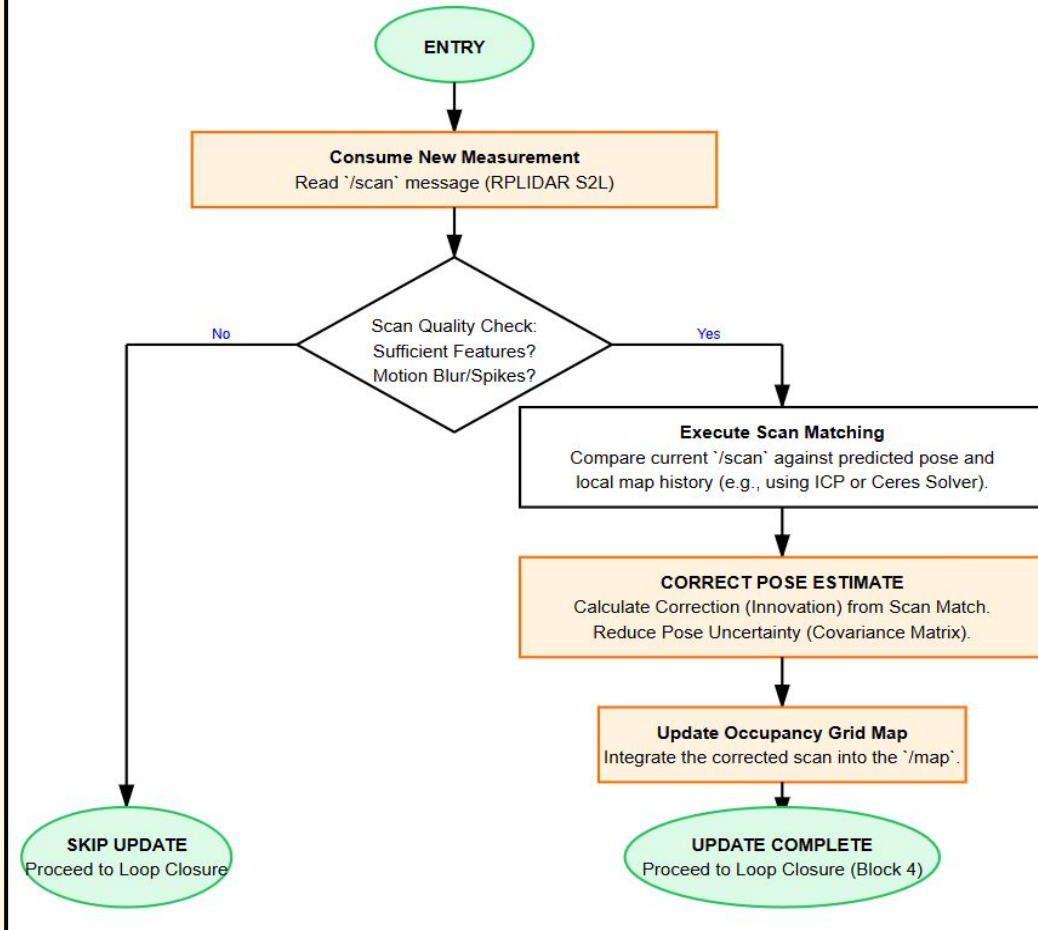
Sub-Flow: READ ALL SENSORS



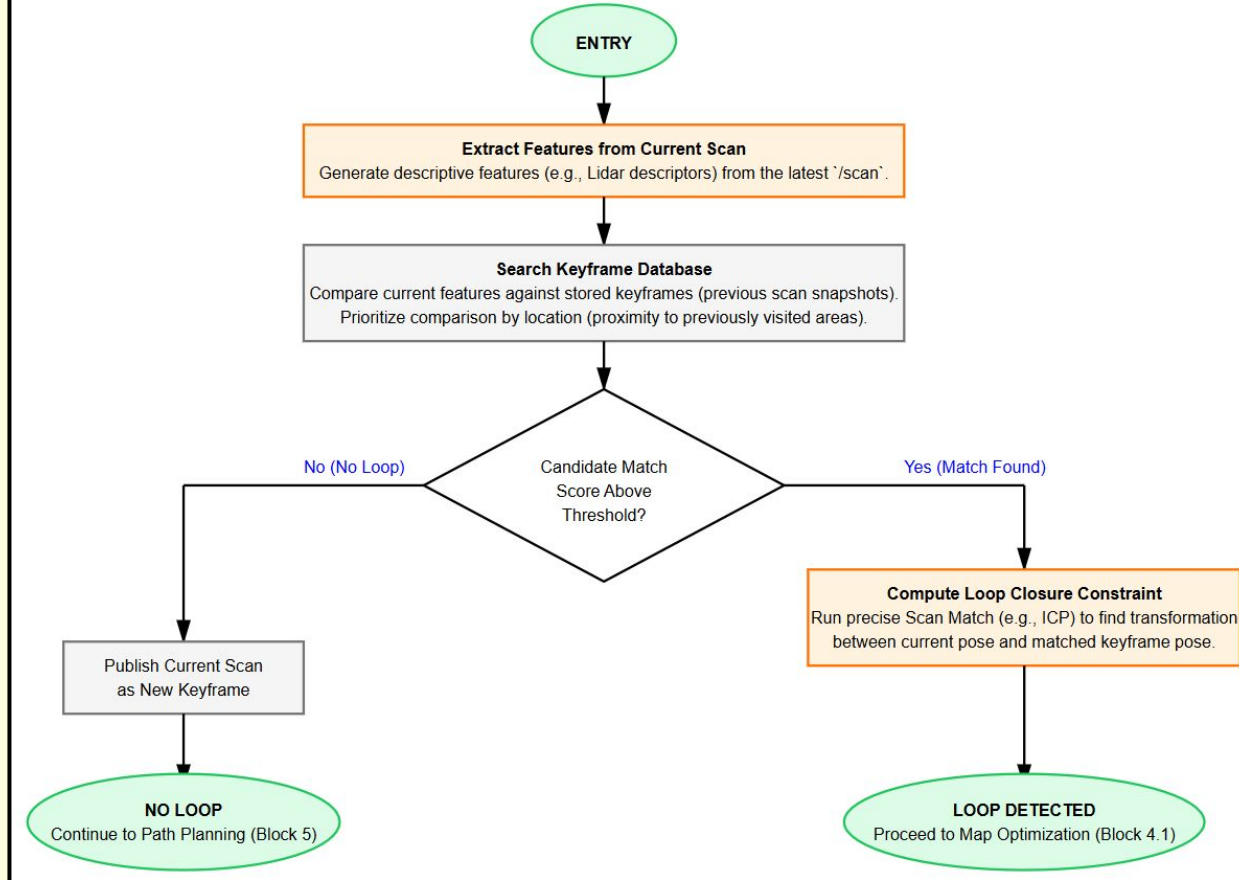
Sub-Flow: SLAM: PREDICTION (Motion Model)



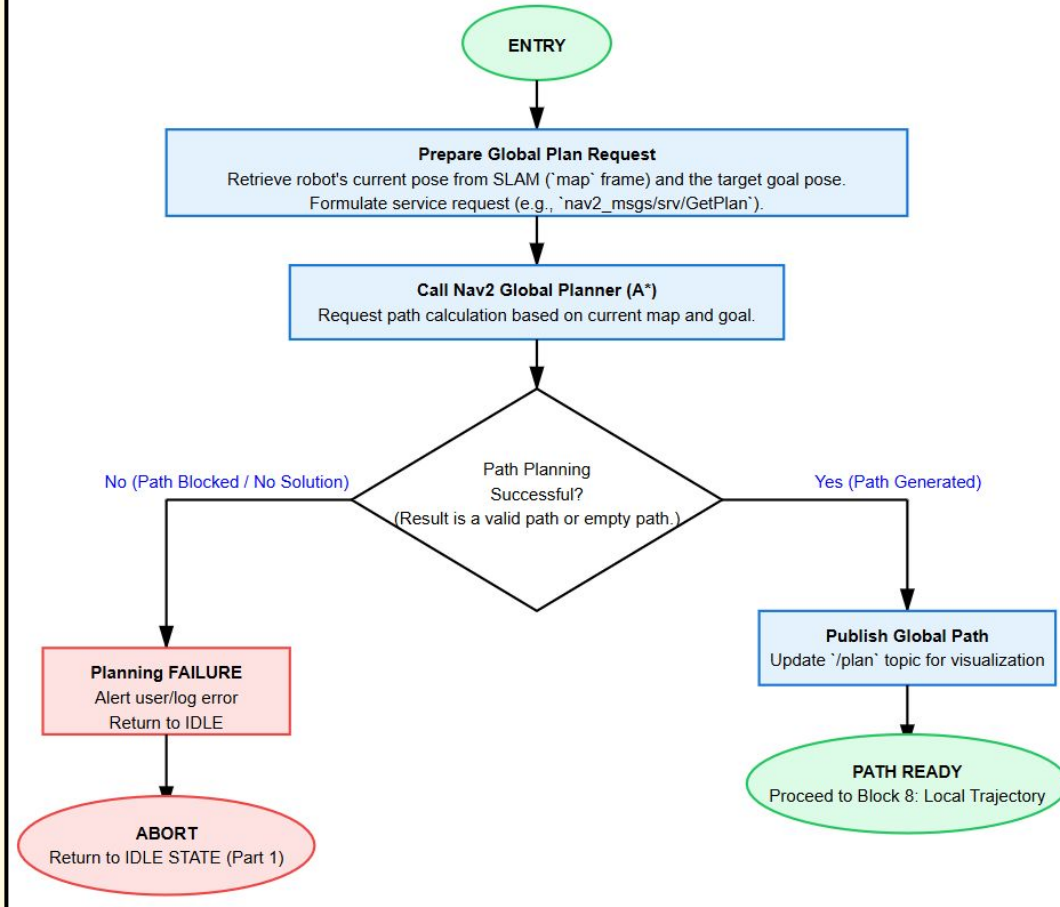
Sub-Flow: SLAM: UPDATE (Correction/Observation Model)



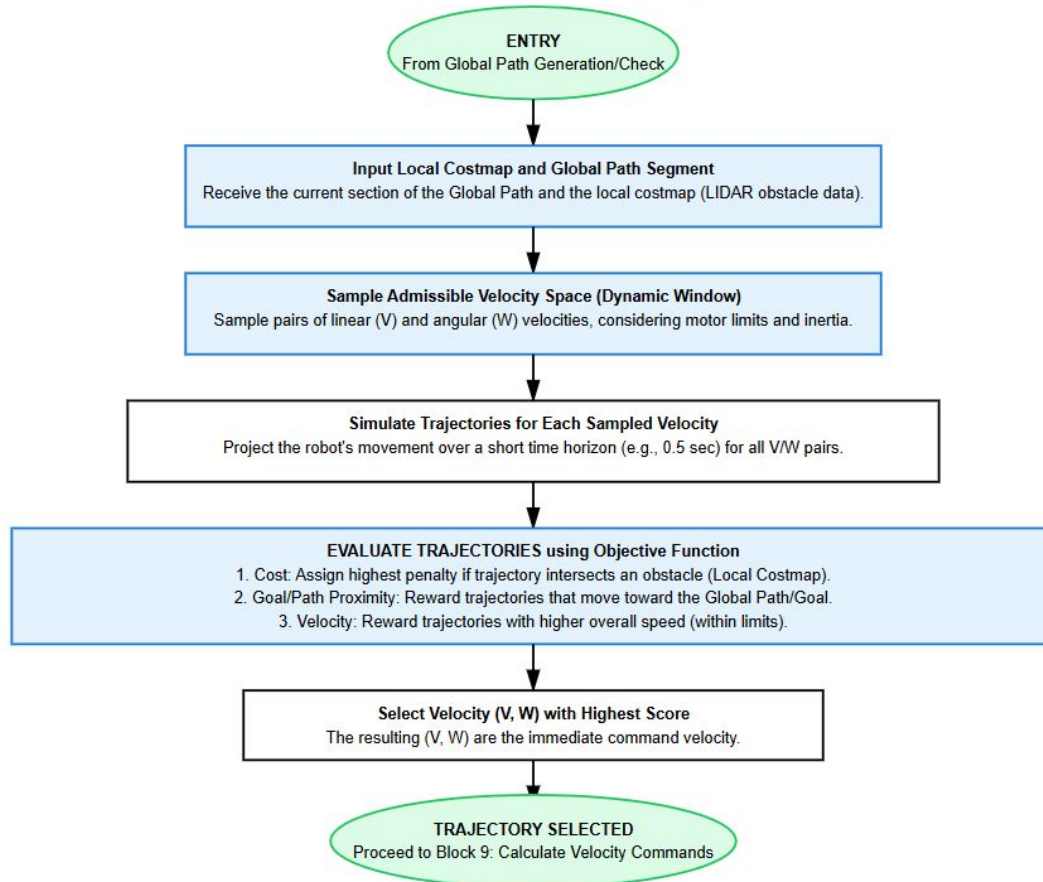
Sub-Flow: Loop Closure Detection



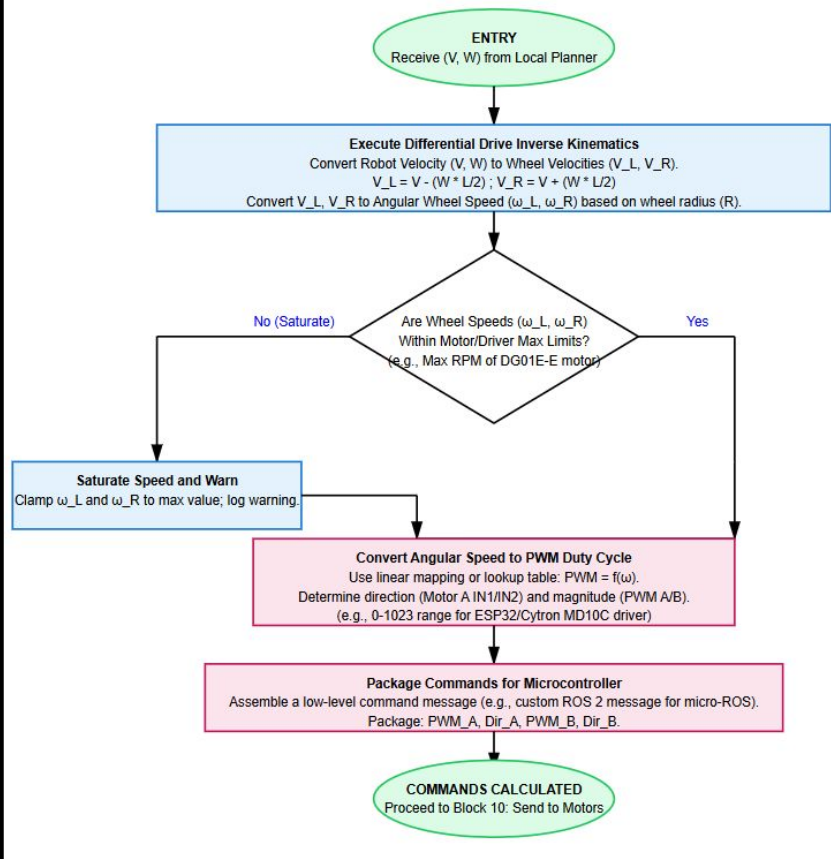
Sub-Flow: Generate Global Path (Global Planner)



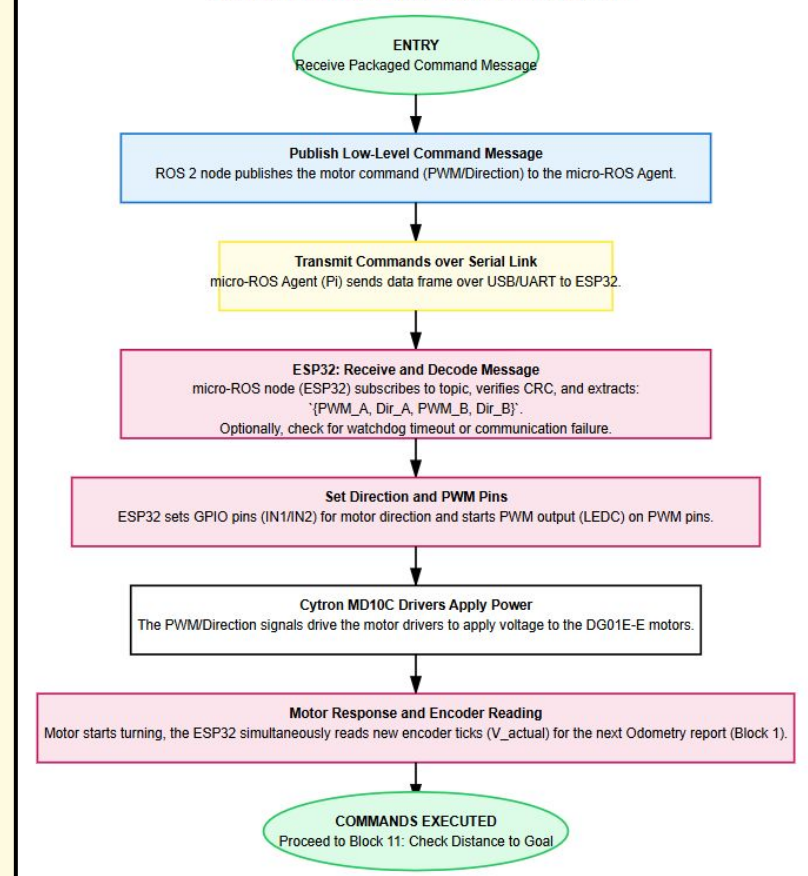
Sub-Flow: Generate Local Trajectory (Dynamic Window Approach - DWA)



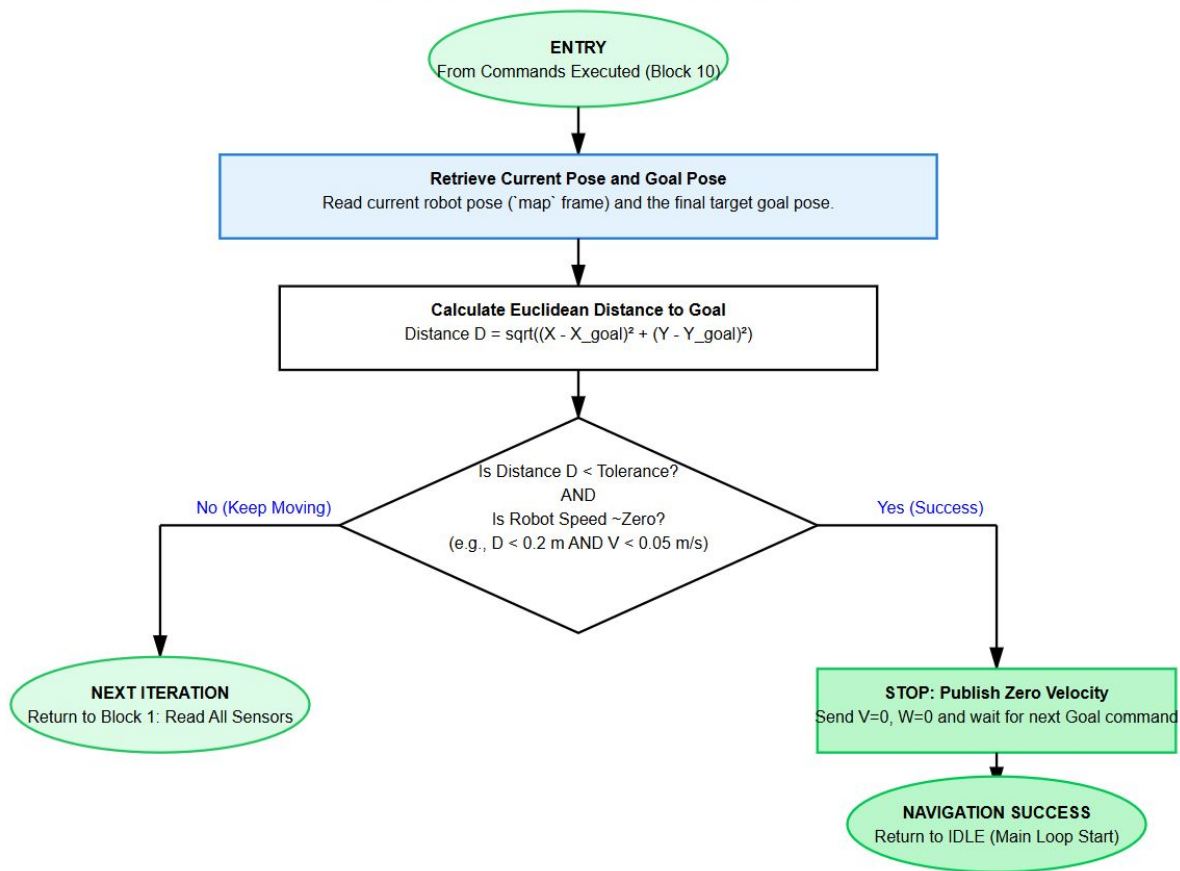
Sub-Flow: Calculate Velocity Commands (Inverse Kinematics)



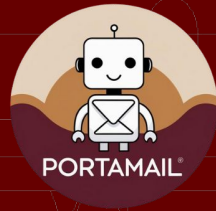
Sub-Flow: Send PWM Commands to Motor Drivers (Execution)

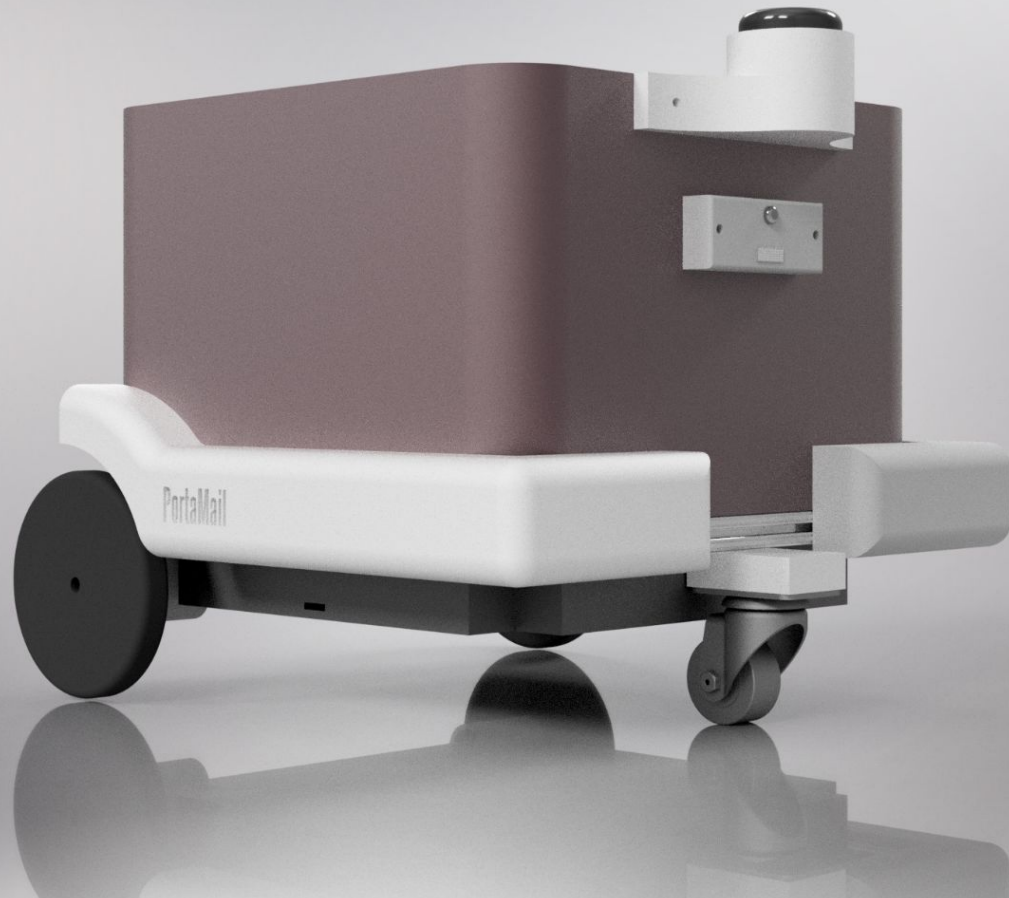


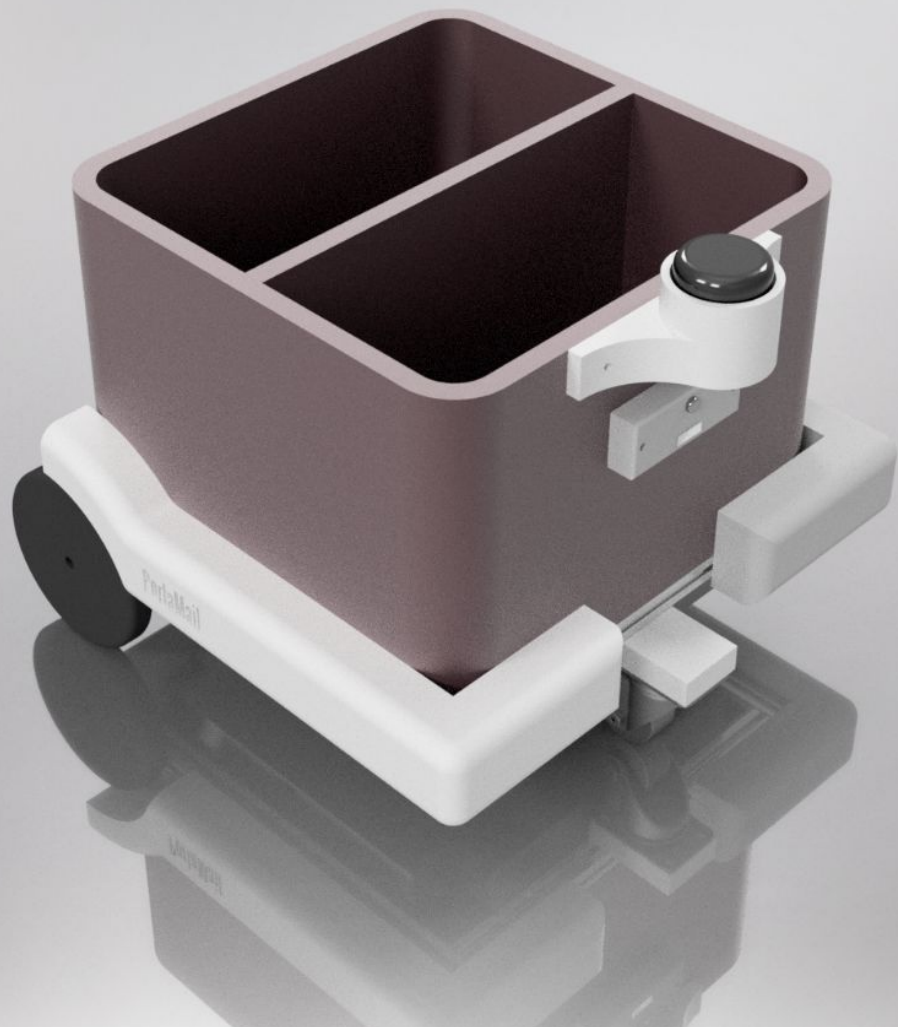
Sub-Flow: Check Distance to Goal (Termination Check)



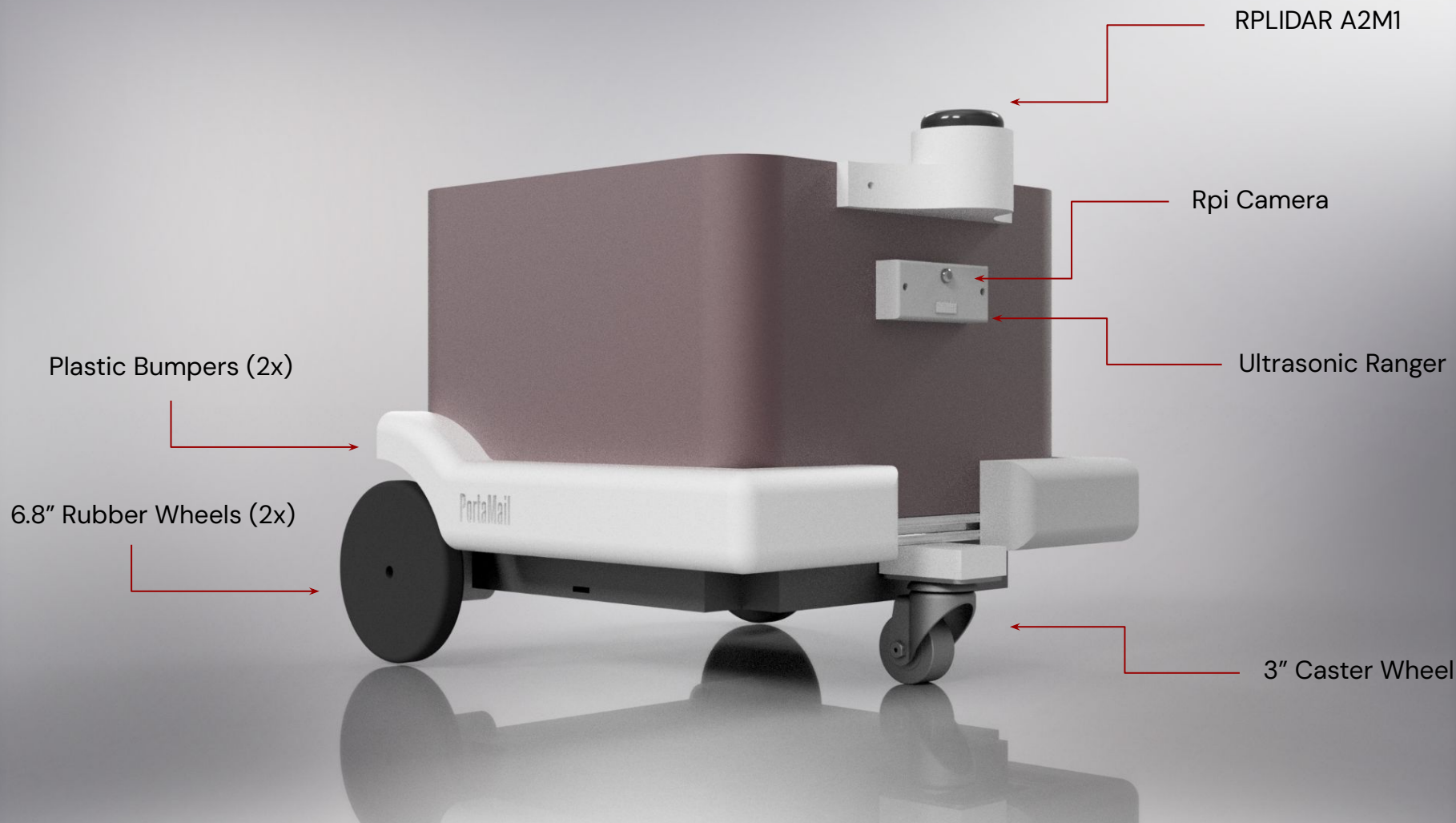
PortaMail Mockups

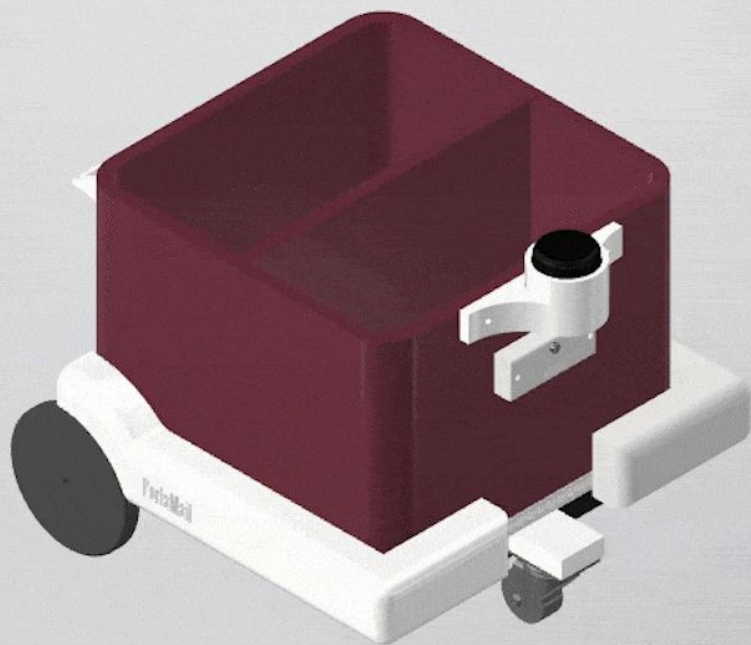




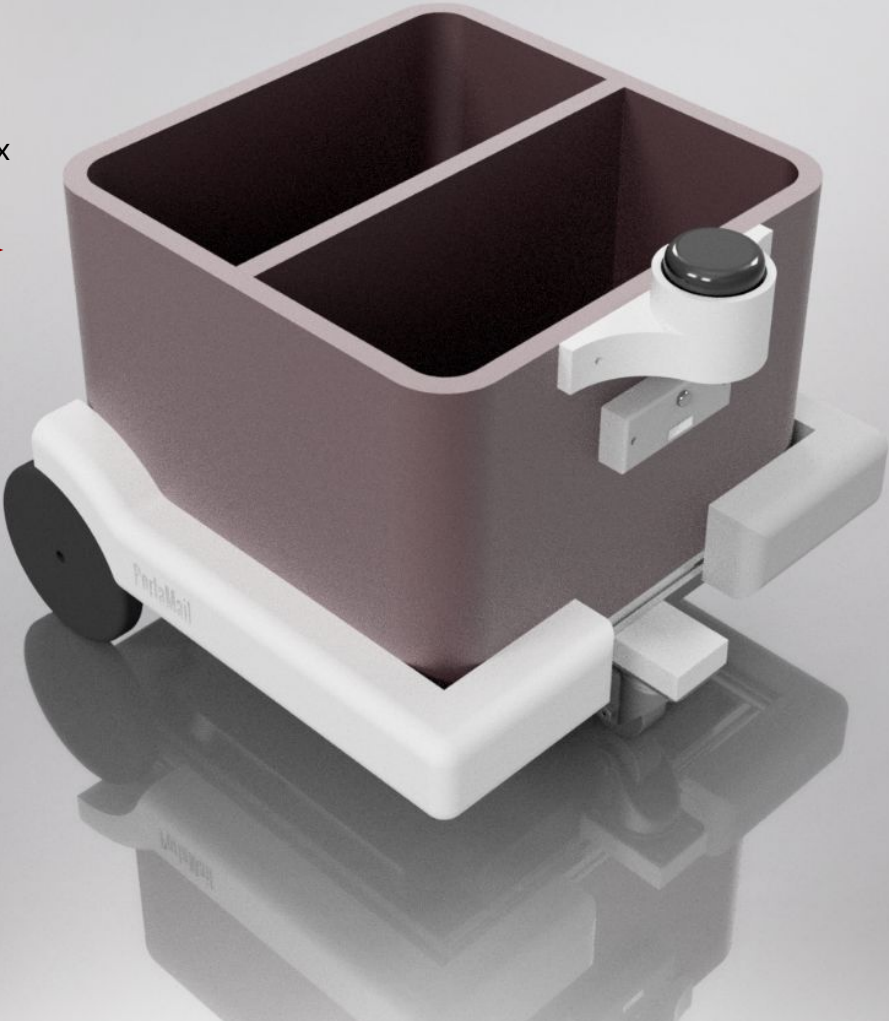
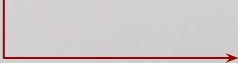






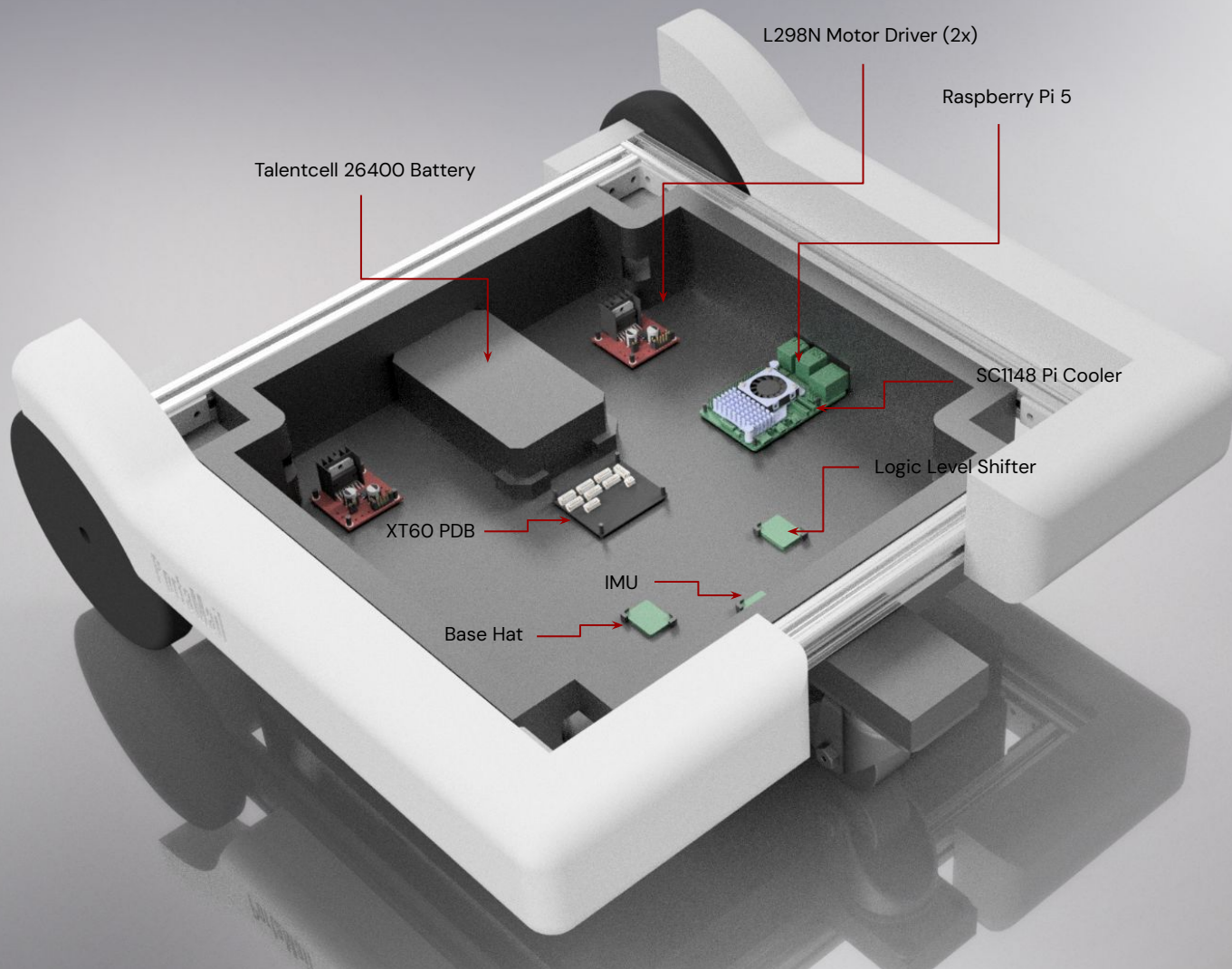


2 Slotted Sliding Mail Holster Box



7" Touch Screen





L298N Motor Driver (2x)

Raspberry Pi 5

Talentcell 26400 Battery

SC1148 Pi Cooler

Logic Level Shifter

XT60 PDB

IMU

Base Hat

Thank You!

